

Alexander of Macedon holding a rocket. From Conrad Kyeser, Bellifortis, *Cod. phil. 63, fol. 11b. (Courtesy of the Niedersächsische Staats- und Universitätsbibliothek, Göttingen.) Kyeser venerated Alexander of Macedon as a commanding military leader from antiquity. Here he shows Alexander holding a powerful rocket inscribed with the mysterious, magically powerful letters* MEUFATON. *Whether or not* MEUFATON *stands for Methone (353 B.C.), a king conquered by the young Alexander, as one scholar has suggested, Kyeser depicts an Alexander endowed with technological and magical power. The representation of a powerful ruler handling military hardware brings together* praxis *and* techne *in a way that would have been contrary to ancient notions of* praxis.

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Power, Patronage, and the Authorship of *Ars*

From Mechanical Know-how to Mechanical Knowledge in the Last Scribal Age

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ABSTRACT

"Power, Patronage, and the Authorship of *Ars*" is a study of books on the mechanical arts that appeared in the fifteenth century from Italy and south Germany. At this time a significant proliferation of such books occurred, some profusely illustrated, others exclusively textual. They include Latin books on military apparatus and other kinds of machines, German-language codices on gunpowder artillery and machines, humanist treatises on painting, sculpture, architecture, and the military arts, and vernacular writings on such subjects by practitioners. The author argues that the patronage that encouraged the creation of such books developed because of a new close alliance between political and military *praxis* and the mechanical arts. She suggests that such authorship elevated the status of the mechanical arts by explicating them in writing, rationalizing them, and associating them with ancient traditions. They were thus prepared for appropriation by the new experimental philosophy of the next era.

MANUSCRIPT BOOKS ON THE MECHANICAL ARTS proliferated rapidly in fifteenth-century Europe, particularly in south Germany and north and central Italy. Despite some important ancient and medieval exemplars, there is simply no precedent for the great number and variety of such codices that appeared from the early years of the

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fifteenth century until its end. Both university-educated humanists and artisan-trained practitioners created treatises, some chiefly pictorial, others primarily textual, on constructive and other processes in the material world. Topics included gunpowder and artillery; machines, such as mills, pumps, fountains, and cranes; machine parts, particularly gears and pulleys; vehicles, including war carriages, carts, cars, and boats; and numerous other devices. Treatises also appeared concerning the ornamental and constructive arts, especially painting, sculpture, and architecture. These books usually have been studied within particular disciplines such as military history, the history of technology, and the history of art and architecture; they have been investigated primarily for their multifarious technical content. This essay focuses, rather, on the authors who created them, on the context of their authorship, and on its cultural implications.[1]

This study is further delimited in that it concerns authorship on the mechanical arts that occurred within the context of scribal rather than print culture. Some of the writings treated here (such as Leon Battista Alberti's *De re aedificatoria*) were later printed, but they were originally composed as handwritten and hand-copied works, either before the invention of the press or outside of its immediate influence. Despite the fact that the printing press was invented in Germany around the mid-fifteenth century, this investigation does not concern printing directly. Yet in its focus on the scribal proliferation of technical books in the era just before printing, it does call for a modification of Elizabeth L. Eisenstein's thesis that posits the press as the most important causal agent for the resurgence of technical literature in early modern Europe.[2]

This investigation concerns authorship on the mechanical arts within the political and cultural context of north Italy and south Germany, and it claims as well to have relevance to the historiography of early modern science. Among other issues, it elucidates the role of the artisan in the development of experimental philosophy. The thesis positing artisanal influence on early modern science originated most significantly in a series of important articles published by Edgar Zilsel in the early 1940s. Zilsel's views have been reiterated and elaborated by Paolo Rossi and, more recently, by J. A. Bennett, among others.[3] In this

[1] See esp. William Crossgrove, *Die deutsche Sachliteratur des Mittelalters* (Bern: Peter Lang, 1994), esp. pp. 103–144; Gerhard Eis, *Mittelalterliche Fachliteratur* (Stuttgart: J. B. Metzlersche, 1962); Bertrand Gille, *Engineers of the Renaissance,* trans. anon. (Cambridge, Mass.: MIT Press, 1966), esp. pp. 55–170; Bert S. Hall, "Der Meister sol auch kennen schreiben und lesen: Writings about Technology ca. 1400–ca. 1600 A.D. and Their Cultural Implications," in *Early Technologies,* Vol. 3, ed. Denise Schmandt-Besserat (Malibu, Calif.: Undena, 1979); Hall, *The Technological Illustrations of the So-Called "Anonymous of the Hussite Wars": Codex Latinus Monacensis 197, Part 1* (Wiesbaden: Ludwig Reichert, 1979), esp. pp. 121–133; and Max Jähns, *Geschichte der Kriegswissenschaften vornehmlich in Deutschland,* Vol. 1: *Altertum, Mittelalter, XV. und XVI. Jahrhundert* (Munich: Oldenbourg, 1889), pp. 241–443. More recently, the best introduction—especially to the Italian material—is Paolo Galluzzi, ed., *Prima di Leonardo: Cultura delle macchine a Siena nel Rinascimento* (Milan: Electa, 1991).

[2] Elizabeth L. Eisenstein, *The Printing Press as an Agent of Change: Communications and Cultural Transformations in Early-Modern Europe,* 2 vols. (Cambridge: Cambridge Univ. Press, 1979), esp. Vol. 2, pp. 520–574. Recent scholarship concerning particular kinds of early modern technical authorship includes William Eamon, *Science and the Secrets of Nature: Books of Secrets in Medieval and Early Modern Culture* (Princeton, N.J.: Princeton Univ. Press, 1994); Pamela O. Long, "The Openness of Knowledge: An Ideal and Its Context in Sixteenth-Century Writings on Mining and Metallurgy," *Technology and Culture,* 1991, *32*:318–355; and Mary J. Voss, "Between the Cannon and the Book: Mathematicians and Military Culture in Sixteenth-Century Italy" (Ph.D. diss., Johns Hopkins Univ., 1994).

[3] See esp. Edgar Zilsel, "The Sociological Roots of Science," *American Journal of Sociology,* 1942, *47*:544–562; and Zilsel, "The Origins of Gilbert's Scientific Method," in *Roots of Scientific Thought: A Cultural Perspective,* ed. Philip P. Wiener and Aaron Noland (New York: Basic, 1957), pp. 219–250. For Zilsel's life and work see Johann Dvořak, *Edgar Zilsel und die Einheit der Erkenntnis* (Vienna: Löcker, 1981). For a collection of Zilsel's essays that includes important biographical and bibliographical scholarship by the editors see Zilsel,

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essay, I extend but also revise the Zilsel thesis in two ways. First, I stress authorship as an important and heretofore neglected modality of artisanal influence. Second, I question Zilsel's view of the artisan as separate in every way from learned and elite groups. Artisanal authorship came about through the patronage of oligarchs and rulers who supported the technological authorship of learned humanists as well. I argue that technical books attracted the patronage of elites because in the fifteenth century the practice and representation of rulership came to be closely associated in particular ways with technological power and the mechanical arts.

Artisan-trained practitioners authored a significant number of books concerning the mechanical arts, although by no means all of them. They represent a select group of artisans, most of whom had turned away from the support of craft guilds to the patronage of elite rulers; they wrote books in part to further that patronage. I suggest that not only artisanal practice per se, but artisanal authorship, influenced the development of thought by changing the status and character of certain mechanical arts.

Some historians focusing on seventeenth-century topics, such as Mario Biagioli and Steven Shapin, have emphasized the low status of the mechanical arts and the efforts of experimental philosophers to dissociate themselves from them. Others have elaborated a picture of more complex interaction.[4] In my view experimental philosophy, given its methodology of testing hypotheses by manipulating mechanical devices, must be said to have appropriated both values and specific knowledge from the mechanical arts. That appropriation was possible because those arts had increased in status and cultural significance in the preceding two centuries. Rather than contest the low status of the mechanical arts within particular local cultures in the late sixteenth and seventeenth centuries, I suggest that there is a complicated prior history concerning those arts that needs further study.

Another issue, underscored recently by Alan Gabbey, involves the transition from the mechanical arts to the science of mechanics. What, he has asked, is the conceptual history that transformed the mechanical arts from "artificial" human construction produced in opposition to nature (the traditional view) to the science of mechanics as a branch of physics devoted to the study of certain aspects of nature?[5] The present study, which is restricted to fifteenth-century manuscript books, obviously does not elucidate the entire

The Social Origins of Modern Science, ed. Diederick Raven, Wolfgang Krohn, and Robert S. Cohen (Boston Studies in the Philosophy of Science) (Dordrecht: Kluwer, forthcoming). Later scholarship that is indebted to Zilsel's work includes Paolo Rossi, *Philosophy, Technology, and the Arts in the Early Modern Era*, trans. Salvator Attanasio, ed. Benjamin Nelson (New York: Harper & Row, Harper Torchbooks, 1970); Pamela O. Long, "The Contribution of Architectural Writers to a 'Scientific' Outlook in the Fifteenth and Sixteenth Centuries," *Journal of Medieval and Renaissance Studies*, 1985, *15*:265–298; J. A. Bennett, "The Mechanics' Philosophy and the Mechanical Philosophy," *History of Science*, 1986, *24*:1–28; and Bennett, "The Challenge of Practical Mathematics," in *Science, Culture, and Popular Belief in Renaissance Europe*, ed. Stephen Pumfrey, Rossi, and Maurice Slawinski (Manchester: Manchester Univ. Press, 1991), pp. 176–190.

[4] Mario Biagioli, "The Social Status of Italian Mathematicians, 1450–1600," *Hist. Sci.*, 1989, *27*:41–95; Biagioli, *Galileo, Courtier: The Practice of Science in the Culture of Absolutism* (Chicago: Univ. Chicago Press, 1993); and Steven Shapin, *A Social History of Truth: Civility and Science in Seventeenth-Century England* (Chicago: Univ. Chicago Press, 1994), esp. pp. 355–407. For views that suggest greater interaction and parity see Alexander Keller, "Mathematics, Mechanics, and the Origins of the Culture of Mechanical Invention," *Minerva*, 1985, *23*:348–361; Bruce T. Moran, "German Prince-Practitioners: Aspects in the Development of Courtly Science, Technology, and Procedures in the Renaissance," *Technol. Cult.*, 1981, *22*:253–274; and Pamela H. Smith, *The Business of Alchemy: Science and Culture in the Holy Roman Empire* (Princeton, N.J.: Princeton Univ. Press, 1994).

[5] Alan Gabbey, "Between *Ars* and *Philosophia naturalis:* Reflections on the Historiography of Early Modern Mechanics," in *Renaissance and Revolution: Humanists, Scholars, Craftsmen, and Natural Philosophers in Early Modern Europe*, ed. J. V. Field and Frank A. J. L. James (Cambridge: Cambridge Univ. Press, 1993), pp. 133–145. See also W. R. Laird, "The Scope of Renaissance Mechanics," *Osiris*, 2nd Ser., 1986, *2*:43–68.

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transformation to which Gabbey alludes. Yet an early part of that process, I suggest, involved the explication and rationalization of certain craft practices in books. To put it another way, the textual and pictorial elaboration of the mechanical arts in codices allowed their transformation from "know-how," available for constructing things in the world, to "knowledge" involving rational or mathematical principles—and taken to be true.

Books on the mechanical arts became useful in patronage relationships because of a newly perceived alliance between technology and military and political power. In the fifteenth century a close affiliation developed between the ancient, separate categories of *praxis,* meaning political and military action, and *techne,* referring to productive and constructive activities in the material world. Military strength and political legitimacy came to be closely associated with construction, technology, and technique. Rulers, princes, and military captains who wanted to consolidate their power achieved legitimation through the remodeling of urban space, the construction and decoration of great palaces, and the creation of painting, sculpture, and other material artifacts. As Christine Smith has put it, such projects were indicative of the "use of the built environment as evidence for the authority of the state."[6]

New political maps were carved by individuals and groups who did not enjoy the legitimacy provided by membership in the traditional noble classes. Urban patriciate elites, including bankers, merchants, and other beneficiaries of late medieval commercial capitalism, created new power configurations. Members of the nobility who did achieve power did so more through military skill than through kinship relationships. In oligarchic republics such as Florence, Venice, Nuremberg, and Augsberg, commercial activity and wealth did not automatically confer political legitimacy. Nevertheless, the German emperors relied on patriciate oligarchs in the imperial cities in their struggle against the traditional nobility. In Italy, men who started their careers as *condottiere,* military captains leading mercenary armies, often achieved autonomous power as princes of city-states. Initially they established control through military force—a consequence of their ability to lead soldiers. Thereafter, like the merchant oligarchs of the republics and the free imperial cities, they sought to augment their authority by other means. They created visible manifestations of power—by reordering urban space, building and decorating great palaces and churches, staging elaborate festivals, and creating courts that became centers of culture and learning.[7]

[6] Christine Smith, *Architecture in the Culture of Early Humanism: Ethics, Aesthetics, and Eloquence, 1400–1470* (New York: Oxford Univ. Press, 1992), p. 50. The distinctions between *episteme* (knowledge of unchanging things), *praxis* (action requiring *phronesis* or judgment in contingent situations), and *techne* (material construction) were emphasized in Aristotle, *Nicomachean Ethics* 6.4–6.5 (1140a–1141a). The analysis and recasting of such distinctions is at the heart of the work of political philosophers such as Hannah Arendt, *The Human Condition* (Chicago: Univ. Chicago Press, 1958); and Jürgen Habermas, *Theory and Practice,* trans. John Viertel (Boston: Beacon, 1973), esp. pp. 41–81, which treats the classical separation of *praxis* and *techne* and its subsequent fusion in the thought of Machievelli. See esp. Joseph Dunne, *Back to the Rough Ground: "Phronesis" and "Techne" in Modern Philosophy and in Aristotle* (Notre Dame, Ind.: Univ. Notre Dame Press, 1993), for an important study of Aristotelian and twentieth-century discussions; and Alasdair MacIntyre, *After Virtue: A Study in Moral Theory,* 2nd ed. (Notre Dame, Ind.: Univ. Notre Dame Press, 1984). Historians of science such as Owen Hannaway and Pamela Smith have much to say about the changing nature of these relationships: Hannaway, *The Chemists and the Word: The Didactic Origins of Chemistry* (Baltimore: Johns Hopkins Univ. Press, 1975); and Smith, *Business of Alchemy* (cit. n. 4), esp. pp. 56–92.

[7] Studies that elucidate relationships between political power and the decorative and constructive arts in the Renaissance include Richard A. Goldthwaite, *The Building of Renaissance Florence: An Economic and Social History* (Baltimore: Johns Hopkins Univ. Press, 1980); Goldthwaite, *Wealth and the Demand for Art in Italy, 1300–1600* (Baltimore: Johns Hopkins Univ. Press, 1993), esp. pp. 212–255; Mary Hollingsworth, *Patronage in Renaissance Italy: From 1400 to the Early Sixteenth Century* (Baltimore: Johns Hopkins Univ. Press, 1994); Thomas DaCosta Kaufmann, "Editor's Statement: Images of Rule: Issues of Interpretation," *Art Journal,* 1989,

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No longer the result of mere craft know-how and skill, the built environment became the material manifestation of rational and mathematical principles that were elaborated in books. Construction projects were carried out with the help of machines that themselves came to be illustrated and explicated on the folios of codices. Similarly, the *praxis* of military leadership came to be closely associated with armaments and techniques—in contrast to most ancient models, in which generalship was perceived to rest on character and qualities of leadership rather than on technology.[8] In this environment technical books, whether lavishly illustrated and thus a visual manifestation of power, or predominantly textual, involving the explication of rational principles, came to be highly appropriate gifts for princes.

Such books necessitated the open display and explication of the technical, craft, and mechanical disciplines that constituted their subject matter. The degree and nature of such openness varied from one author to another. Some authors candidly explained and illustrated aspects of their subject while suggesting that further secrets might be revealed if employment or patronage were forthcoming. Others wrote in an unambiguously open fashion. In either case, the openness of fifteenth-century books on the mechanical arts constitutes a striking contrast to the well-documented secrecy of late medieval craft guilds.[9]

The argument I have outlined can be demonstrated only by careful attention to specific authors, the books that they created, and the contexts within which they undertook authorship. Fifteenth-century books on the mechanical arts can conveniently be separated into four groups: illustrated Latin treatises from the first half of the century; German treatises, composed throughout the century, that primarily although not exclusively treat gunpowder artillery; humanist Latin writings; and, finally, diverse treatises by practitioners, mostly Italian, mostly from the second half of the fifteenth century. It will be seen

48:119–122; Joanna Woods-Marsden, "Images of Castles in the Renaissance: Symbols of 'Signoria'/Symbols of Tyranny," *ibid.*, pp. 130–137; Kaufmann, *Court, Cloister, and City: The Art and Culture of Central Europe, 1450–1800* (Chicago: Univ. Chicago Press, 1995); Bram Kempers, *Painting, Power, and Patronage: The Rise of the Professional Artist in the Italian Renaissance,* trans. Beverley Jackson (London: Allen Lane, Penguin, 1993); F. W. Kent and Patricia Simons with J. C. Eade, eds., *Patronage, Art, and Society in Renaissance Italy* (Canberra: Humanities Research Centre, 1987); Lauro Martines, *Power and Imagination: City-States in Renaissance Italy* (New York: Random House, Vintage, 1980); Charles M. Rosenberg, ed., *Art and Politics in Late Medieval and Early Renaissance Italy: 1250–1500* (Notre Dame, Ind.: Univ. Notre Dame Press, 1990); Randolph Starn and Loren Partridge, *Arts of Power: Three Halls of State in Italy, 1300–1600* (Berkeley/Los Angeles: Univ. California Press, 1992); Roy Strong, *Art and Power: Renaissance Festivals, 1450–1650* (Berkeley/Los Angeles: Univ. California Press, 1984); and David Thomson, *Renaissance Architecture: Critics, Patrons, Luxury* (Manchester: Manchester Univ. Press, 1993).

[8] See Pamela O. Long and Alex Roland, "Military Secrecy in Antiquity and Early Medieval Europe: A Critical Reassessment," *History and Technology,* 1994, *11*:259–290, which suggests that the openness of much ancient military culture (with the exception of imminent battle situations) is based on the association of military success with the character of the military leader rather than with military technology.

[9] For secrecy and intellectual property attitudes within medieval craft contexts see Pamela O. Long, "Invention, Authorship, 'Intellectual Property,' and the Origin of Patents: Notes toward a Conceptual History," *Technol. Cult.,* 1991, *32*:846–884. Openness and secrecy within fifteenth-century technical authorship are complex issues that will be treated at length in my book in progress: Long, "Openness, Secrecy, Authorship, Ownership: Studies in Premodern and Early Modern Practical, Technical, and Knowledge Traditions." My point here is that the context of authorship within patronage relationships involved at least some requisite display and explication of technological material and led to the open explication of the mechanical arts in manuscript books. Here and elsewhere, issues of openness and secrecy are influenced closely by context. See William Eamon, "From the Secrets of Nature to Public Knowledge," in *Reappraisals of the Scientific Revolution,* ed. David C. Lindberg and Robert S. Westman (Cambridge: Cambridge Univ. Press, 1990), pp. 333–365; and Roger Hahn, "The Meaning of the Mechanistic Age," in *The Boundaries of Humanity: Humans, Animals, Machines,* ed. James J. Sheehan and Morton Sosna (Berkeley/Los Angeles: Univ. California Press, 1991), pp. 142–157, both of which associate openness more broadly with developments in science (such as "mechanization of the world picture") or with technological developments, such as printing.

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that the books themselves often contain crucial evidence concerning patronage or hoped-for patronage. Moreover, the explication of machines and instruments is at times closely connected with assumptions about political and military power and the representations of that power. Finally, the mechanical arts are rationalized in a variety of ways and newly associated with learning, the liberal arts, and classical antiquity.

BETWEEN EMPIRE AND CITY-STATE: LATIN TECHNICAL AUTHORSHIP IN THE EARLY FIFTEENTH CENTURY

The first group of writings to be considered comprises Latin technical treatises from the early fifteenth century. The authors of these books, Conrad Kyeser (1366–1405), Giovanni Fontana (ca. 1393–ca. 1455), and Mariano Taccola (1382–1453), explicated numerous kinds of machines and weapons both in writing and pictorially.[10] The three men worked in the contiguous regions of north and central Italy and south Germany. The area was divided linguistically and encompassed a multitude of changing political and military territories, but it also contained common features and was characterized by porous boundaries. The German emperors had long-standing interests in Italian territories, certain components of which were formally if not effectively part of the empire. Italian princes received patronage from the emperor and fought for him as *condottiere.* Mercenary soldiers traveled back and forth between Italian and German regions; most gunners in north Italian armies in the early fifteenth century were German. Other facilitators of interchange between the two regions were the church, commerce, and banking. In important ways, the three authors shared a single political and military culture.[11]

Conrad Kyeser, the son of South German burghers, authored *Bellifortis* [*Strong in War*] between 1402 and 1405 while in exile in his home town of Eichstätt. A strikingly illustrated treatise written in Latin verse, *Bellifortis* is the earliest fifteenth-century technical book from the German territories. Kyeser was exiled in 1402 when Sigismund, king of Hungary and of Germany, defeated Prague, imprisoned his half-brother, the Bohemian king Wen-

[10] Kyeser, who had worked as a physician in the imperial army, wrote his military treatise, *Bellifortis,* between 1402 and 1405; see Conrad Kyeser aus Eichstätt, *Bellifortis,* ed. and trans. Götz Quarg (Düsseldorf: Verlag des Vereins Deutscher Ingenieure, 1967). Fontana's treatise, *Bellicorum instrumentorum liber,* treats military machines, fountains, and other mechanical devices and includes Latin text that is partially encrypted. See Eugenio Battisti and Giuseppa Saccaro Battisti, *Le macchine cifrate di Giovanni Fontana* (Turin: Arcadia, 1984). Mariano Taccola was a notary from Siena who authored *De ingeneis* in the 1420s and 1430s and *De machinis* in the 1440s. See Mariano Taccola, *De ingeneis: Liber primus leonis, Liber secundus draconis, addenda; Books I and II, On Engines, and Addenda (The Notebook),* ed. and trans. Gustina Scaglia, Frank D. Prager, and Ulrich Montag, 2 vols. (Wiesbaden: Ludwig Reichert, 1984); Prager and Scaglia, *Mariano Taccola and His Book "De ingeneis"* (Cambridge, Mass.: MIT Press, 1972); and Taccola, *Liber tertius de ingeneis ac edifitiis non usitatis,* ed. J. H. Beck (Milan: Il Polifilo, 1969). For *De machinis* see Taccola, *De machinis: The Engineering Treatise of 1449,* ed. and trans. Scaglia, 2 vols. (Wiesbaden: Ludwig Reichert, 1971); and Taccola, *De rebus militaribus (De machinis, 1449),* ed. Eberhard Knobloch (Baden-Baden: Valentin Koerner, 1984).

[11] For the older literature see Paolo Brezzi, "I comuni cittadini italiani e l'impero medioevale," in *Nuove questioni di storia medioevale* (Milan: Marzorati, 1969), pp. 177–207. See also Heinz Angermeier, "Die Sforza und das Reich," in *Gli Sforza a Milano e in Lombardia e i loro rapporti con gli stati italiani ed europei (1450–1535)* (Convegno internazionale, Milano, 18–21 maggio 1981) (Milan: Istituto Editoriale Cisalpino–La Goliardica, 1982), pp. 165–191; Hermann Kellenbenz, "Oberdeutschland und Mailand zur Zeit der Sforza," *ibid.,* pp. 193–227; and M. E. Mallett and J. R. Hale, *The Military Organization of a Renaissance State: Venice c. 1400 to 1617* (Cambridge: Cambridge Univ. Press, 1984), for much information about the relationship of one city-state, Venice, to the empire, including Venetian recruitment of German soldiers (pp. 21, 79). For German soldiers in Italy see Mallett, *Mercenaries and Their Masters: Warfare in Renaissance Italy* (London: Bodley Head, 1974), esp. pp. 27–36; Karl H. Schäfer, *Deutsche Ritter und Edelknechte in Italien während des 14. Jahrhunderts* (Paderborn: Ferdinand Schöningh, 1911); and Marino Viganò, ed., *Architetti e ingegneri militari italiani all'estero dal XV al XVIII secolo* (Livorno: Sillabe, 1994).

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ceslas, and sent the court, presumably including Kyeser, into exile. Banished to his hometown of Eichstätt in Bavaria, Kyeser wrote his treatise in the last years of his life. The manuscript, of which there are at least fourteen exemplars, was illustrated by illuminators from the Prague scriptorium who were passing through Eichstätt after their own expulsion.[12]

Bellifortis is the response of a bitterly angry military physician to the imperial politics of his time. Kyeser had been present at Nicopolis on that "impious day" in 1396 when the Ottoman Turks had precipitated the flight "with unheard of audacity of Prince Sigismund, hermaphroditic Hungarian king," and his imperial army. He was a victim in the struggle between Sigismund and Wenceslas, both sons of the Luxembourg emperor Charles IV. Kyeser dedicated his treatise to the emperor Ruprecht (ruled 1400–1410) from the Palatinate (Pfalz), a weak emperor chosen by the electors after their unprecedented deposal of Wenceslas in 1400.[13]

In *Bellifortis* Kyeser offers technological strength to the emperor Ruprecht, advertising the efficacy of astrology, sorcery, mechanical technology, and weaponry. The lavish illustrations of the treatise depict cannon, rockets, chariots, a counterweight trebuchet, battering rams, mobile bridges, ships, mills, scaling ladders, incendiary devices, crossbows, and instruments of torture—weapons, machines, and devices both old and new. The large, expensive format, the striking illustrations, and the text in Latin verse all suggest that the emperor Ruprecht to whom the treatise was dedicated was also representative of the intended audience. Kyeser wrote a prince's book, not an engineer's.[14]

[12] Kyeser, *Bellifortis,* ed. and trans. Quarg (cit. n. 10), is an edition and German translation of the work that includes extensive notes and a facsimile reproduction of a manuscript at Göttingen dated 1405 (Niedersächsische Staats- und Universitätsbibliothek, MS philos. 63 cim.). However, see the critical review of Quarg's edition by Hermann Heimpel in *Göttingische Gelehrte Anzeigen,* 1971, *223*:115–148. The identity of the *Bellifortis* Kyeser with a "Master Conrad of Eichstätt" who authored medical tracts is uncertain. Lynn White, Jr., "Kyeser's 'Bellifortis': The First Technological Treatise of the Fifteenth Century," *Technol. Cult.,* 1969, *10*:436–441, on p. 438 n. 2, identified Kyeser as the "Master Conrad of Eichstätt" who composed a collection of medical recipes in an early fifteenth-century manuscript associated with the court of Wenceslas in Prague. But Günter Kallinich and Karin Figala, "Konrad von Eichstätt, eine Arztpersönlichkeit des deutschen Mittelalters," *Sudhoffs Archiv,* 1969, *52*:341–346, argued that the author of medical texts was a well-known physician from Eichstätt who lived in the early fourteenth century.

[13] Kyeser, *Bellifortis,* ed. and trans. Quarg, Vol. 1, p. 4 (fol. 3a): "diem impium," "inaudite audacie principis Sygismundi Regis ermifroditi ungarie." In a later passage (Vol. 1, p. 53, fol. 85a) Kyeser reiterates that the Turks had defeated "Sigismund, a deserter and madman, deceitful good-for-nothing [Sygmundum perfugum atque furibundum/Fallacem nequam]." The disadvantages of a weak emperor had become apparent after the electors had watched Ruprecht lead an embarrassingly incompetent imperial army in and out of Italy in 1402. The electors reverted back to the Luxembourg line in 1410, choosing the far more powerful, if less malleable, Sigismund. Thereafter, Sigismund became king of Bohemia in 1419 (after the death of Wenceslas) and king of the Lombards in 1431. He was crowned Holy Roman Emperor in 1433. For Ruprecht see Aug. Thorbecke, "Ruprecht III," in *Allgemeine deutsche Biographie* (Leipzig: Duncker & Humblot, 1889), Vol. 29, pp. 716–726; for his 1402 military adventure in Italy see Brian Pullan, *A History of Early Renaissance Italy: From the Mid-Thirteenth to the Mid-Fifteenth Century* (London: Allen Lane, 1973), pp. 234–235. For Sigismund's career see esp. János Bak, "The Late Medieval Period, 1382–1526," in *A History of Hungary,* ed. Peter F. Sugar *et al.* (Bloomington: Indiana Univ. Press, 1990), pp. 54–64; Wilhelm Baum, *Kaiser Sigismund: Hus, Konstanz und Türkenkriege* (Graz: Styria, 1993); Friedrich Bernward Fahlbusch, *Städte und Königtum im Frühen 15. Jahrhundert: Ein Beitrag zur Geschichte Sigmunds von Luxemburg* (Cologne: Böhlau, 1983); Heinrich Koller, "Sigismund, 1410–1437," in *Kaisergestalten des Mittelalters,* ed. Helmut Beumann (Munich: C. H. Beck, 1984), pp. 277–300, 375–376; Elemér Mályusz, *Kaiser Sigismund in Ungarn, 1387–1437* (Budapest: Akadémiai Kiadó, 1990); and Sabine Wefers, *Das politische System Kaiser Sigmunds* (Stuttgart: Franz Steiner, 1989). See also Bertalan Kéry, *Kaiser Sigismund: Ikonographie* (Vienna: Anton Schroll, 1972), a thorough study of the many portraits and other visual images of Sigismund. For the Battle of Nicopolis see Aziz Suryal Atiya, *The Crusade of Nicopolis* (London: Methuen, 1934).

[14] I refer to the illustrations in the facsimile edition—Kyeser, *Bellifortis,* ed. and trans. Quarg—of one particular manuscript (Göttingen, Niedersächsische Staats- und Universitätsbibliothek, MS philos. 63 cim.). The illustra-

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As would befit a prince, the technology of *Bellifortis* includes a full panoply of magical and mechanical weapons. Kyeser enhances the prestige of that technology by tying it to both learning and classical antiquity. He insists upon the relevance of letters to military strength: "The ambition of Caesar [i.e., Sigismund] is foolish, for art is worth gold / And letters rule the shield and sword of the soldier / Letters alone govern the chasuble and the robe / All things are under letters, with which the One Almighty rules."[15] Kyeser praises the strong soldiers of Germany and simultaneously applauds her liberal and mechanical arts: "just as the sky shines with stars, Germany shines forth with liberal disciplines, is embellished with mechanics, and adorned with diverse arts."[16]

Kyeser's hero is Alexander of Macedon, conqueror of the east, whose triumph stands in stark contrast to Sigismund's eastern debacle at Nicopolis. Yet his portrait of Alexander as technologically efficacious in his own person is antithetical to ancient notions of military leadership. In *Bellifortis* Alexander stands with a rocket-like weapon inscribed with the mysterious letters *MEUFATON* (see Frontispiece). The accompanying verses explain the efficacy of the weapon against all enemies. Elsewhere, Kyeser reports that Alexander is the inventor of a huge war carriage (see Figure 1). He thus represents Alexander's power as rooted in technical competence that is both mechanical and magical.[17] Kyeser's Alexander brings together the categories of *praxis* and *techne*. In striking contrast to ancient mores, he is a military leader armed with magical, mechanical, and incendiary devices.

Kyeser presents himself as a dying man in 1405, the date inscribed at the end of his dedication to Ruprecht. His own portrait appears near the end of the treatise, accompanied by a self-composed funeral ode and epitaph, a list of rulers and princes with whom he had been associated (Wenceslas, Sigismund, and Francis of Carrara in Padua, among others), and a prayer for his about-to-be departed soul (see Figure 2). He includes verses for the arts that have informed his treatise—the seven liberal arts and also geomancy, theology, philosophy, law, canon law, physics, alchemy, the theurgical arts, and the military arts. He represents himself as famous and loved by all and as an expert on military matters whose *Bellifortis* will scatter armies. He will be widely mourned, by nobles, the wealthy, the poor, and even animals—quadrupeds, tripeds, birds, worms.[18]

Nothing in Giovanni Fontana's writings corresponds to the bizarre finale of Kyeser's treatise. Yet Fontana shared with his German counterpart both a delight in mechanical contrivances and the assumption that power is associated with technological efficacy. Fontana first took up technical authorship at the University of Padua, where he wrote three small treatises—two concerning clocks and one treating measurement from a distance.

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tions in other copies (e.g., New York Public Library, Spencer MS 58) are not in every case identical to those of the Göttingen codex.

[15] Kyeser, *Bellifortis*, ed. and trans. Quarg, Vol. 1, p. 2 and n. 1 (fol. 2a): "Caesaris ambicio desipit, nam ars valet aurum / Et littera clippeum militis gerit atque mucronem. / Casulam cum stola gubernat littera sola / Litteris cuncta subsunt, agla quibus imperat unus." Quarg suggests that "*agla*" is an old Hebraic name for "Almighty," derived from the first four letters of an ancient Jewish prayer and used by Kyeser to display his linguistic knowledge. "*Stola*," in line 3, can refer to a royal robe, in which case the line suggests that letters govern both priestly and royal realms.

[16] *Ibid.*, Vol. 1, p. 3 (fol. 2b): "sicud celum ornatur sideribus, sic alemania prefulget disciplinis liberalibus, honestatur mechaniis diversisque artibus adornatur."

[17] *Ibid.*, Vol. 1, p. 15, and Vol. 2 (fol. 11b), for Alexander holding the rocket; and Vol. 1, p. 17, and Vol. 2 (fol. 16a) for the war carriage. Quarg suggests that *MEUFATON* refers to Philip of Macedonia's conquest of Methone in 353 B.C., during which conflict the Macedonian king lost an eye. What Kyeser makes clear is the almost magical properties of the weapon, which will cause enemies to flee. Concerning the war carriage, Kyeser writes: "Hoc instrumentum ab allexandro repertum [this instrument was invented by Alexander]."

[18] *Ibid.*, Vol. 1, pp. 100–105 (fols. 135a–139a).

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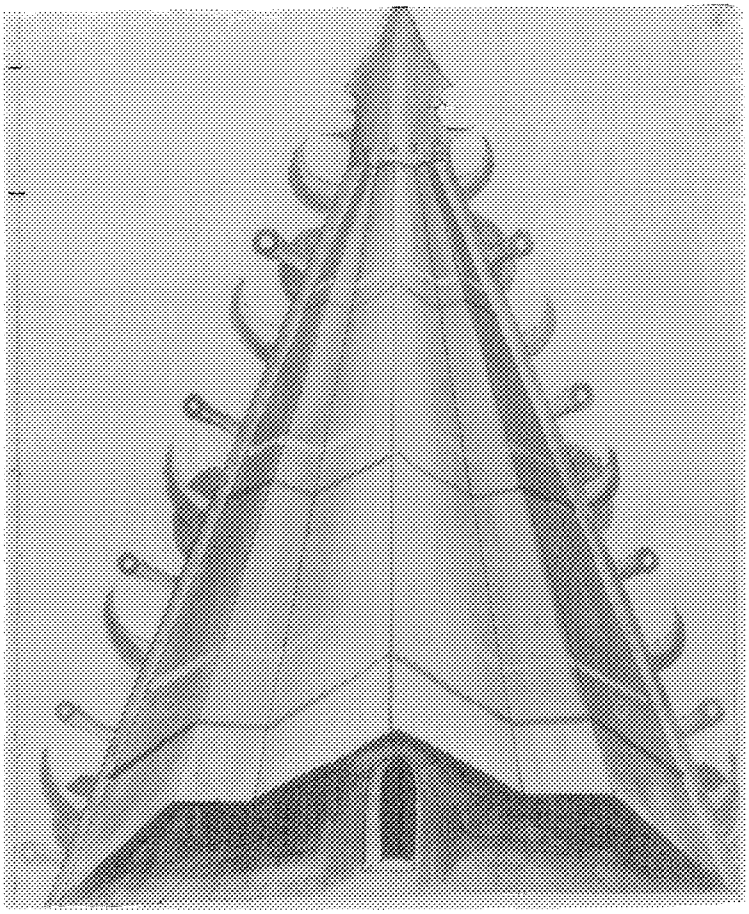


Figure 1. *A giant tank invented by Alexander of Macedon. From Conrad Kyeser, Bellifortis, Cod. phil. 63, fol. 16a. (Courtesy of the Niedersächsische Staats- und Universitätsbibliothek, Göttingen.) Kyeser again emphasizes the technological skill of a powerful ruler with his depiction of this gigantic tank, which he claims was invented by Alexander of Macedon. To view the tank coherently, push the ram at the top down and flip the round wheels on the sides underneath. The soldiers enter by the door in front. The vehicle's defenses include a combination of scythes and cannon placed in alternating positions between the wheels, which are (we are told) hidden. Thus can the army safely approach the enemy and destroy it. The tradition extolling Alexander here enters a new phase, as the Macedonian king becomes the inventor of large-scale military hardware. The illustration beautifully evokes the combination of ancient warfare (scythes) with modern weaponry (cannon).*

After receiving his medical degree in 1421, evidence suggests, he worked as a military physician and was based, for a time at least, with the Venetian army in Brescia.[19]

[19] For a discussion of Fontana's life and writings see Marshall Clagett, "The Life and Works of Giovanni Fontana," *Annali dell'Istituto e Museo di Storia della Scienza di Firenze*, 1976, *1*:5–28; and Battisti and Saccaro Battisti, *Macchine cifrate di Fontana* (cit. n. 10), pp. 7–52. His three early treatises, bound together in the university library in Bologna (Bibliotheca universitaria, 2705), are *De horologio aqueo* (fols. 53r–84v), on water clocks; *Nova compositio horologii quod ex pulverum casu consistit* (fols. 1r–52v), on sand clocks; and *De pisce, cane et volucre* (from fol. 85r), on measurement from a distance. Fontana's illustrations were clearly influenced by Giovanni da Dondi, whose treatise on the astrarium he must have seen in Padua. See Jane Andrews Aiken,

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Figure 2. *Portrait of Conrad Kyeser, author of the* Bellifortis. *From Conrad Kyeser,* Bellifortis, *Cod. phil. 63, fol. 139a. (Courtesy of the Niedersächsische Staats- und Universitätsbibliothek, Göttingen.) This remarkable portrait depicts Kyeser wearing a green shirt, the color of physicians and of those born under the sign of Jupiter (including Alexander of Macedon). The portrait appears in the codex accompanied by the author's self-composed epitaph and funeral ode. In contrast to the magical and technological powers emanating from the* Bellifortis, *Kyeser presents himself (with a deeply scarred face, presumably displaying a battle wound) as a dying man. The blank space that he left for his date of death was never filled in.*

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The military context of such a career involves the early fifteenth-century expansion of Venetian rule to the *terra ferma* in the form of direct military control and the creation of a standing army to protect the newly acquired territory. Venetian conflicts included the successful war against the Carrara of Padua in 1404–1405, the defense against Hungarian invasions, prompted by Sigismund's expansionist policies, in 1411–1413 and 1418–1421, and the wars against Milan between 1426 and 1454. Venetian military expansion required the hiring of captains-general, a move undertaken with great trepidation by the Venetian senate. Fontana's presence in Brescia is presupposed by his documented role in transmitting a message from the senate to the captain-general of the Venetian army, the ill-fated Francesco Carmagnola. Although Fontana's whereabouts for a number of years are undocumented, he records in a treatise of the 1440s some of the details of his life as a physician in Udine, an outpost of the Venetian army.[20]

It seems probable that Fontana wrote his *Bellicorum instrumentorum liber* in the 1420s and intended it for Carmagnola. The Latin treatise, which concerns military machines, fountains, and other mechanical devices, is written partially in cryptographic code. It consists of seventy folio sheets and close to 140 drawings of numerous kinds of machines and devices: fountains and pumps, magic lanterns, machines for lifting and transporting heavy weights, clocks, defensive and offensive devices to protect forts and castles, battering rams, incendiary and other kinds of weapons, alchemical furnaces, magic lanterns, measuring instruments, a mechanical devil and witch, towers, locks, keys, and scaling ladders, to name a few. Although the illustrations are often self-explanatory, most sheets also contain textual explication that begins in alphabetic Latin and then shifts to encryption. Fontana uses identical encryption in a later work, the *Secretum de thesauro experimentorum,* that treats experiments and natural philosophy.[21]

The encryption of the *Bellicorum instrumentorum liber* can be seen as a display of writing technique involving codification rather than as a means of concealing technical information. In a cogent analysis, the editors of the facsimile edition conclude that the single existing manuscript of the work and the manuscript of Fontana's *Secretum* were both produced by scribes. They hypothesize that Fontana developed the code when he wrote the *Secretum* and that he then instructed scribes to use it to copy the earlier *Bellicorum instrumentorum liber.* In the latter, encrypted text usually appears directly under alphabetic Latin on each page, regardless of the content. On one sheet, for example, Fontana illustrates a fountain and in alphabetic Latin exclaims upon his love for fountains and

"Truth in Images: From the Technical Drawings of Ibn Al-Razzaz Al-Jazari, Campanus of Novara, and Giovanni de' Dondi to the Perspective Projection of Leon Battista Alberti," *Viator,* 1994, *25*:325–359; and Carlo Maccagni, "Le scienze nello studio di Padova e nel Veneto," in *Storia della cultura Veneta dal primo quattrocento al Concilio di Trento,* Vol. 3, ed. Girolamo Arnaldi and Manlio Pastore Stocchi (Vicenza: Neri Pozza, 1981), pp. 135–171, esp. pp. 142–147. Fontana's other writings include *Tractatus de trigono balistario* (Oxford, Bodl. libr. canon. misc. 47), a treatise on measurement using the triangle, composed in Udine in 1440; and a natural history that exists in a printed edition—*Liber Pompilii Azali Placentini De omnibus rebus naturalibus* . . . (Venice: Ottaviano Scoto, 1544). He also wrote a work, presumed lost, on perspective that he dedicated to the Venetian painter Jacopo Bellini. Battisti and Saccaro Battisti hypothesize that it may be the partially extant perspectival treatise attributed to Toscanelli in Alessandro Parronchi, "Le fonti di Paolo Uccello, II: I 'filosofi,' " *Paragone,* 1957, no. 95, pp. 3–33; see Battisti and Saccaro Battisti, *Macchine cifrate di Fontana,* pp. 18–24.

[20] For Venetian military matters in the early fifteenth century see Mallett and Hale, *Military Organization of a Renaissance State* (cit. n. 11), esp. pp. 20–43, upon which this account depends. Battisti and Saccaro Battisti, *Macchine cifrate di Fontana,* pp. 27–28, notes Fontana's mission to Brescia. Fontana's view of the captain-general Carmagnola's execution by order of the senate in 1432 can perhaps be inferred from his recording of the evil omens that he saw beforehand.

[21] See Battisti and Saccaro Battisti, *Macchine cifrate di Fontana,* pp. 35–37 (concerning the encryption), 141–158 (for the *Secretum de thesauro*). The manuscript is in Paris, Bibliothèque Nationale, Lat. nouv. acq. 635.

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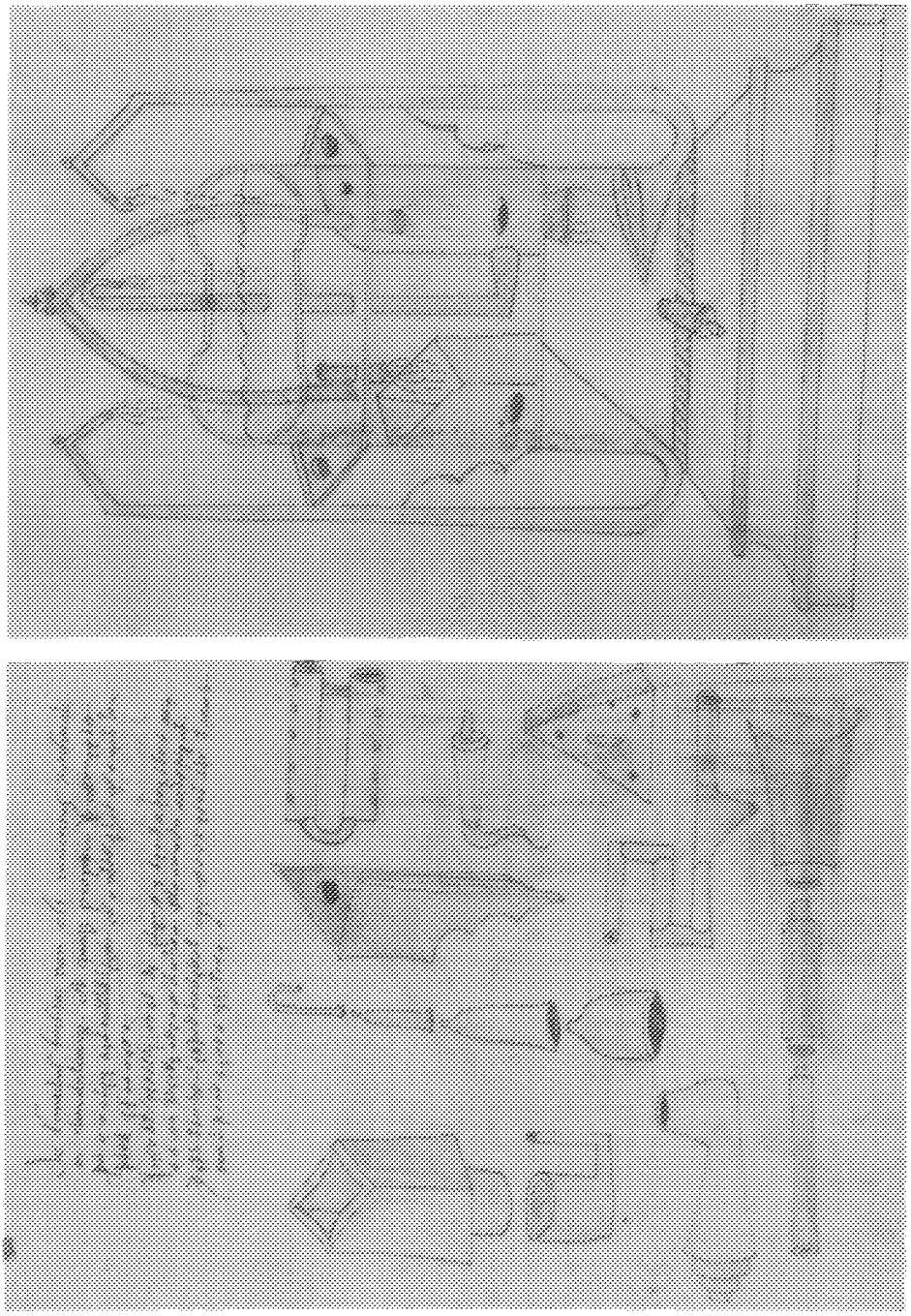


Figure 3. *Parts of a fountain* (left) *and a transparent view of the assembled fountain* (right). *From Giovanni Fontana,* Bellicorum instrumentorum liber, *Cod. icon. 242, fols. 62v and 63r. (Courtesy of the Bayerische Staatsbibliotheque, Munich.) Here Fontana shows the parts of a fountain and how they are assembled into a working fountain. He records in alphabetic Latin that he delights in the study of fountains, whereas in the encrypted Latin he assures the reader that everything has been explained clearly. This example suggests that Fontana did not use encryption to hide technical information.*

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his great skill and ingenuity in inventing them (see Figure 3). The encrypted section includes the following: "All parts of the Fountain, and the complete figure of this thus are depicted clearly so that you can understand with facility."[22] The secrecy embodied in the encryption is unrelated to technological detail. As the editors of the work suggest, Fontana was interested in the idea of the code itself rather than in hiding technological secrets per se.

Fontana's contemporary Mariano Taccola, from Siena, authored two pictorial machine books: *De ingeneis,* created in the 1420s and early 1430s, and *De machinis,* composed in the Veneto in the 1440s. Although they are by no means identical, both works are filled with drawings of constructions and machines—cannon, trebuchets, incendiary weapons, winches, pumps, mills, gears, cranes, and ships, among many others. Most sheets also contain explanatory Latin text.[23]

Mariano came from a middle level of Sienese society—his father was a wine dealer, and his wife, Madonna Nanna, was the daughter of a leather merchant. Evidence suggests that his early education included the surprising combination of craft practice and notarial training, including Latin. In 1424 he became a *camarius,* or secretary, of the Domus Sapientae, a hospital and student lodging that served as a temporary residence for foreign visitors. Guests included some of the entourage of Sigismund during the emperor's nine-month Sienese stay in 1432, an interlude necessitated by negotiations with the papacy concerning the imperial coronation. Later, Mariano became a *stimatore* for Siena, one who tallied completed work at a building site, assigned wages, and perhaps estimated the amounts of required materials. In 1441 he was appointed *viaio,* a superintendent of streets, fountains, and bridges.[24]

On one level, Mariano's technical treatises speak to the concerns and problems of the

[22] Battisti and Saccaro Battisti, *Macchine cifrate di Fontana,* p. 96 (fol. 62v of manuscript): "Partes omnes fontis, et figura eius completa ita clare depingantur, ut cum levitate intelligas." See Frank D. Prager, "Fontana on Fountains: Venetian Hydraulics of 1418," *Physis,* 1971, *13*:341–360, for a detailed study of Fontana's fountains. Battisti and Saccaro Battisti's conclusion that the single extant copy was produced by scribes is made on the basis of analysis of mistakes, which are of the sort that would have been made by a copyist rather than by the author. They call Fontana's cryptographic system a simple, rational cipher based on signs without letters or numbers. Fontana had devised a "complete and organic system of signs" that was equivalent to and could substitute for the conventional alphabet. They conclude that Fontana's cryptographic system responded "not so much to the criterion of secrecy as to an awareness of a conscious symbolism of signs and their systems." He was interested in "a system of significance" rather than a system "of the occultization of significance." Battisti and Saccaro Battisti, *Macchine cifrate di Fontana,* pp. 36–37 (my translation of the Italian).

[23] For a facsimile of the first two books (Munich, Bayerische Staatsbibliothek, Codex latinus monacensis 197, Pt. II) with transcription, translation, and notes see Taccola, *De ingeneis,* ed. and trans. Scaglia *et al.* (cit. n. 10). See also Prager and Scaglia, *Taccola and His Book "De ingeneis"* (cit. n. 10), which contains reproductions of the sheets, transcription, and translation of books 3 and 4. For a facsimile edition of books 3 and 4 see Taccola, *Liber tertius de ingeneis,* ed. Beck (cit. n. 10), which includes transcriptions of the documents pertaining to Taccola as well. For *De machinis* see Taccola, *De machinis,* ed. and trans. Scaglia (cit. n. 10), which includes a facsimile edition of Munich, Bayerische Staatsbibliothek, Codex latinus monacensis 28 800; see also Taccola, *De rebus militaribus (De machinis, 1449),* ed. Knobloch (cit. n. 10), which includes a facsimile of Paris, Bibliothèque Nationale, Codex parisinus latinus 7239. An excellent introduction to Taccola and fifteenth-century Sienese culture is Paolo Galluzzi, "Le macchine senesi: Ricerca antiquaria, spirito di innovazione e cultura del territorio," in *Prima di Leonardo,* ed. Galluzzi (cit. n. 1), pp. 15–44.

[24] The evidence for his background and training includes his commissions to carve heads for the choir stalls in 1408 and to produce sculptures in 1441, both for the Sienese cathedral; and his nomination in 1417 as an applicant for admission to the Sienese Guild of Judges and Notaries. For a summary of Taccola's career see Taccola, *De ingeneis,* ed. and trans. Scaglia *et al.,* Vol. 1, pp. 11–15; and Prager and Scaglia, *Taccola and His Book "De ingeneis,"* pp. 3–21. Most of the relevant documents are published in Gaetano Milanesi, *Documenti per la storia dell' arte senese,* Vol. 2: *Secoli XV e XVI* (1854; rpt., Holland: Davaco, 1969), pp. 284–286. For the role of the *stimatore* see Nicholas Adams, "The Life and Times of Pietro dell'Abaco, a Renaissance Estimator from Siena (Active 1457–1486)," *Zeitschrift für Kunstgeschichte,* 1985, *48*:384–395.

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Sienese state. They explicate numerous aspects of ships, harbors, and naval warfare, reflecting Siena's efforts to control Tuscan rivers and to develop a commercial port at Talamone on the Tuscan Sea. Taccola's many depictions of pumps, dams, siphons, aqueducts, and other water-controlling devices constitute a response to the perennial Sienese problem of a water supply insufficient for the needs of the population and local industries, especially leather and textile manufacture. His multifarious depictions of cannon, firearms, cannon lifting and transport devices, and weaponry of various sorts suggest a response to Sienese military problems that included the frequent ravaging of the *contada,* or countryside, by marauding mercenary companies and the constant pressure of Florentine expansionist efforts (see Figure 4).[25]

If Taccola's authorship developed out of Sienese concerns, he also used it in a quest for imperial patronage. In 1432 he presented Sigismund with a copy of books 3 and 4 of the *De ingeneis* that included two portraits of the emperor.[26] Taccola specifies that he desires a position in Sigismund's Hungarian court. Pictorially, he has Saint Dorothy recommend him as a master of machines for waterworks in the Hungarian regions (he has drawn her pointing to the relevant text), and he includes authorship (involving a compilation of the emperor's "places") as one of the services he can perform. Although he probably never made it to Hungary, there is indirect evidence that Taccola worked in the Veneto and northern Italy in the late 1430s and 1440s and that he wrote his later treatise *De machinis,* completed in 1449, in the Veneto as well. For example, a copy was made for the Venetian captain-general Bartolomeo Colleoni; his authorial efforts may also have been prompted by renewed hope for imperial patronage, this time from the early Hapsburg emperor Frederick III, whose coronation in Rome was being discussed in Italy in 1447.[27]

De machinis is a pictorial treatise, written as an aid to "fighting the infidels." It is divided into ten sections or books and contains more than two hundred illustrations, many of which

[25] For an introduction to Sienese history and culture in the fourteenth and fifteenth centuries see esp. Mario Ascheri, *Siena nel Rinascimento: Istituzioni e sistema politico* (Siena: Edizioni "Il Leccio," 1985); Duccio Balestracci and Gabriella Piccinni, *Siena nel trecento: Assetto urbano e strutture edilizie* (Florence: Cooperativa Libraria Univ. Studi Florentini, 1977); William M. Bowsky, *A Medieval Italian Commune: Siena under the Nine, 1287–1355* (Berkeley/Los Angeles: Univ. California Press, 1981); Bowsky, "Siena," in *Dictionary of the Middle Ages* (New York: Scribner's, 1987); David L. Hicks, "The Sienese State in the Renaissance," in *From the Renaissance to the Counter-Reformation: Essays in Honor of Garrett Mattingly,* ed. Charles H. Carter (New York: Random House, 1965), pp. 75–94; and Judith Hook, *Siena: A City and Its History* (London: Hamish Hamilton, 1979). The Sienese aqueducts are the subject of a dissertation-in-progress by Michael P. Kucher (Univ. Delaware). See also Balestracci *et al., I bottini medievali di Siena* (Siena: Alsaba, 1993). The aim of the harbor was to rival those at Pisa and Genoa and to assist in importing grain from Sicily in the face of chronic food shortages.

[26] The manuscripts of books 3 and 4 of *De ingeneis* are in Florence, Biblioteca Nazionale, Palat. 766, fols. 27–57 and 58–76, respectively. See Prager and Scaglia, *Taccola and His Book "De ingeneis"* (cit. n. 10), pp. 70–71, for the portraits and the accompanying text, which appears on the verso along with an illustration of the measurement of the height of a tower. For the portrait of Sigismund see James H. Beck, "The Historical 'Taccola' and Emperor Sigismund in Siena," *Art Bulletin,* 1968, *50*:309–320; Lionello G. Boccia, "L'armamento defensivo nei trattati di Mariano di Iacopo detto il Taccola," in *Prima di Leonardo,* ed. Galluzzi (cit. n. 1), pp. 47–56, esp. pp. 52–53; and Gustina Scaglia, "An Allegorical Portrait of Emperor Sigismund by Mariano Taccola of Siena," *Journal of the Warburg and Courtauld Institutes,* 1968, *31*:428–434.

[27] Prager and Scaglia, *Taccola and His Book "De ingeneis,"* pp. 15–16 (for the titles that he received as a result of his presentation), 144–145 (for his recommendation by St. Dorothy). Although Sigismund seems never to have granted Taccola's wish to join the court in Hungary, he did name him *familiares* in 1432, and in the next year in Mantua, on his return to German territories, he granted him and some other Sienese citizens the title *comites palatini.* For (inconclusive) evidence concerning Taccola's authorship of *De machinis* in the Veneto see Milanesi, *Documenti,* Vol. 2 (cit. n. 24), pp. 299–300; Prager and Scaglia, *Taccola and His Book "De ingeneis,"* pp. 16–20; and esp. Taccola, *De machinis,* ed. and trans. Scaglia (cit. n. 10), Vol. 1, pp. 19–23, 29–58 (for a detailed discussion of the manuscripts).

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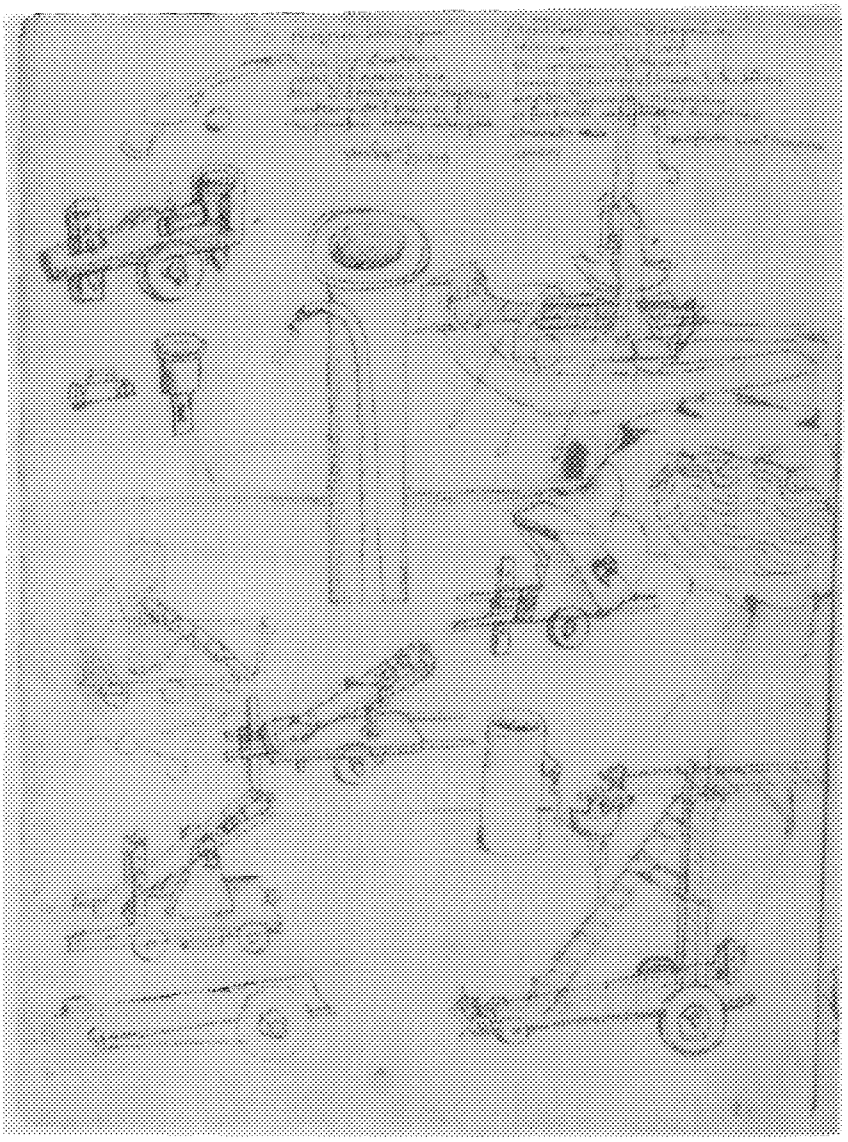


Figure 4. *A sheet of Taccola's* De ingeneis *depicting cannon, a pump, and other devices. From Mariano Taccola,* De ingeneis, *Codex Latinus Monacensis 197, Pt. II, fol. 52v. (Courtesy of the Bayerische Staatsbibliothek, Munich.) This sheet is one of several hundred in Taccola's notebook. It includes sketches of machines and devices central to both Taccola's own interests and those of Siena—military equipment, including mortars, cannon, cannon carriages, and scaling ladders, and hydraulic equipment such as a pump and a syphon.*

are explained textually. The table of contents represents an early attempt to rationalize types of machines into coherent groups, such as "Bridges and Structures on Water," "Bombards and Guns," and "Weights Raised Higher and Lowered Down."[28]

[28] Taccola, *De machinis,* ed. and trans. Scaglia, Vol. 1, p. 18 (for the list of subjects), 59–60 (for Taccola's detailed statement about the uses of his book for fighting the infidels and "bad Christians" who try to usurp the land and power of other Christians).

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In this, his later treatise, Taccola calls himself the "Archimedes of Siena." Archimedes' fame as an inventor of military machines was widespread in the fifteenth century, especially through the account in Plutarch's *Life of Marcellus.* Yet Plutarch's Archimedes explicitly declined to write down his military and mechanical inventions because of the vulgarity of such a utilitarian activity.[29] In striking contrast, Taccola, the Sienese Archimedes, embraces technical authorship, presenting a great array of machines with both pictorial and textual explication; he leaves no doubt that they constitute both the proper subject of rational thought and the essential instruments of civic success and military victory.

Although the treatises of Kyeser, Fontana, and Taccola exhibit important differences, they also share a number of characteristics. They are written in Latin, are lavishly illustrated, and display a variety of mechanical contrivances and weapons in lucid detail. They are dedicated to rulers or, in some cases, seem to have been written for rulers or military captains. They directly associate military and political power with technological power in a new way. They enhance the status of mechanical disciplines by explicating them in Latin and by associating them with antiquity and with learning.

GERMAN BOOKS ON GUNPOWDER AND ARTILLERY

Alongside these Latin books, numerous German-language treatises, primarily concerning gunpowder artillery and machines, appeared beginning in the early decades of the fifteenth century and continuing until its end. Although the specific authors of these books are often unknown, clearly many of them were practitioners in the burgeoning field of gunpowder artillery. Their books contain a wealth of technical information, such as how to make gunpowder and how to load cannon. Yet they also attempt to rationalize their practice by treating the causes of particular phenomena, and they treat ethical issues such as the desirable attitudes and deportment of both practitioners and patrons. These books transmitted technical knowledge concerning gunpowder and artillery. Yet they also elevated such practices by making them rational, by explicating them in writing, and by holding practitioner and patron to a set of ethical norms.[30]

An example from the early fifteenth century, Cgm 600 in the Bayerische Staatsbibliothek in Munich, is a small book of twenty-two sheets that treats the manufacture of gunpowder

[29] *Ibid.*, pp. 59–60: "However, I, Ser Marianus Taccola, also called Archimedes of great and magnificent Siena, have not designed these engines, machines and weapons that they may operate against Christians, but have invented, composed and designed them that they may go against the infidel and barbarian people." For Plutarch's account see *Plutarch's Lives,* 11 vols., Vol. 5: *Agesilaus and Pompey, Pelopidas and Marcellus,* trans. Bernadotte Perrin (London: William Heinemann, 1917), pp. 479–481 ("Life of Marcellus" 17.4). For Archimedes in the fifteenth century see W. R. Laird, "Archimedes among the Humanists," *Isis,* 1991, *82*:629–638; and D. L. Simms, "Archimedes and the Invention of Artillery and Gunpowder," *Technol. Cult.,* 1987, *28*:67–79.

[30] Over a hundred individual examples of fifteenth-century manuscript military treatises of German provenance are extant, including multiple copies of particular texts. The most complete recent list is Hall, *Technological Illustrations* (cit. n. 1), pp. 121–133; see also Jähns, *Geschichte der Kriegswissenschaften,* Vol. 1 (cit. n. 1), pp. 241–443. For gunpowder artillery and other military technologies that form the technological context of this literature see esp. Kelly DeVries, *Medieval Military Technology* (Peterborough, Ont.: Broadview, 1992), pp. 143–168; Bert S. Hall, *Weapons and Warfare in Renaissance Europe: Gunpowder, Technology, and Tactics* (Baltimore: Johns Hopkins Univ. Press, 1997); Gerhard W. Kramer, *Berthold Schwarz: Chemie und Waffentechnik im 15. Jahrhundert* (Munich: Oldenbourg, 1995); and Volker Schmidtchen, *Bombarden, Befestigungen, Büchsenmeister: Von den ersten Mauerbrechern des Spätmittelalters zur Belagerungsartillerie der Renaissance: Eine Studie zur Entwicklung der Militärtechnik* (Düsseldorf: Droste, 1977). See also Erich Egg, *Der Tiroler Geschützguss: 1400–1600* (Innsbruck: Universitätsverlag Wagner, 1961); and Egg, *Das Handwerk der Uhr- und der Büchsenmacher in Tirol* (Innsbruck: Universitätsverlag Wagner, 1982), esp. pp. 184–201.

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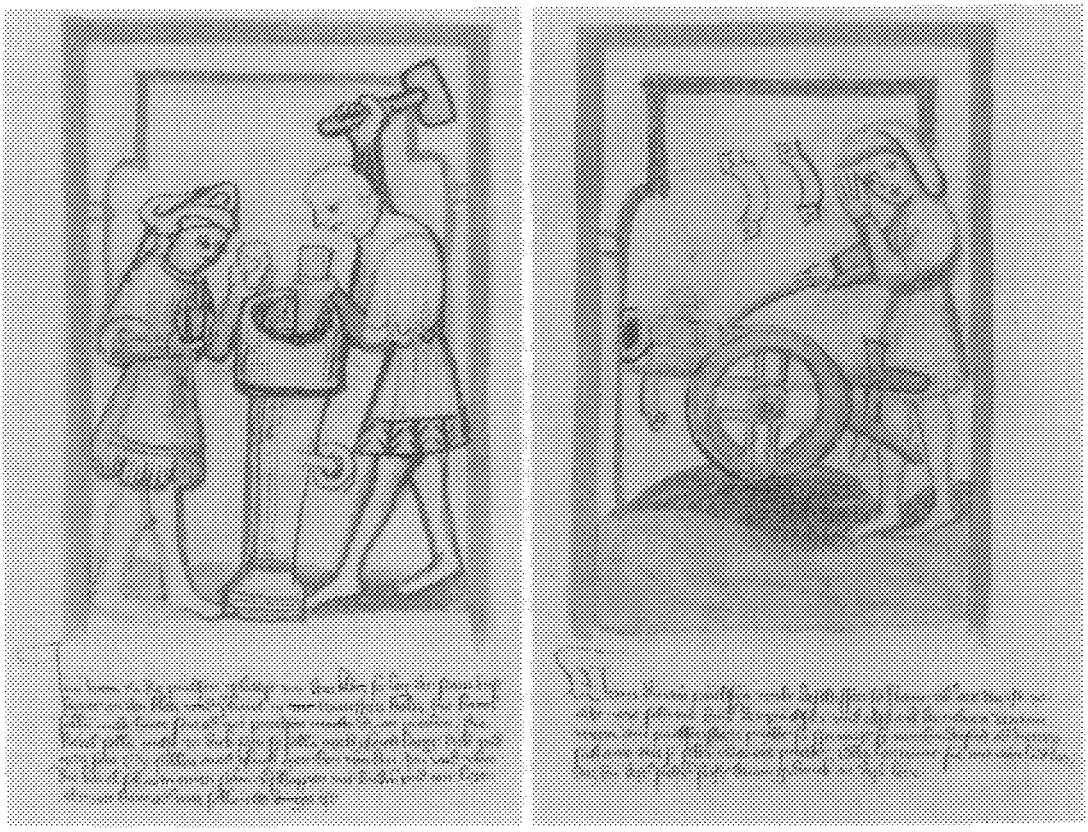


Figure 5. Loading a cannon barrel and igniting the powder. From Cod. germ. 600. (Courtesy of the Bayerische Staatsbibliothek, Munich.) The illustration on the left shows two gunners, depicted in an early fifteenth-century book on gunpowder artillery, loading a cannon barrel. As the text explains, one is hammering in wooden wedges between the stone ball and the interior surface of the barrel. The other holds containers with other substances to be poured in—loam, oakum (part of flax), or hay. On the right, a gunner is igniting the cannon. The text warns of the danger of the barrel exploding. The gunner should stand back and ignite the powder with a long rod. The clamps that secure the barrel to the carriage are shown but not mentioned in the text. Notice the serious expressions, the industrious demeanor, and the simple clothing of the artisans, each of whom is depicted with his own distinctive clothing, suggesting his individuality.

and loading and firing cannon. On most pages a pictorial image dominates. For example, on one page instructions are given concerning how to load the barrel of the cannon. Another illustration (see Figure 5) shows a gunner igniting a cannon, a highly dangerous task: sometimes the weapon itself exploded rather than discharging the ammunition. The text cautions the gunner to stand some distance away and to use a rod when lighting the powder.[31] Cgm 600 transmits technical information but also presents images of practitioners deeply absorbed in technical procedures. Each of the human figures depicted in the woodcuts is distinguishable from the others—each has a distinct physiognomy, wears his

[31] Munich, Bayerische Staatsbibliothek, Cgm 600. For a detailed description see Karin Schneider, ed., *Die deutschen Handschriften der Bayerischen Staatsbibliothek, München,* new ed., Vol. 5, Pt. 4 (Wiesbaden: Otto Harrassowitz, 1978), p. 227. See also Wilhelm Hassenstein, *Das Feuerwerkbuch von 1420* (Munich: Verlag der Deutschen Technik, 1941), pp. 48, 56.

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own particular clothes, and sports his own hat or hairstyle. Some work in pairs, some alone. All are dressed plainly and clearly belong to the artisan class. All are busy at work, with serious demeanor, carrying out the procedures described in the text below. Although authored anonymously, Cgm 600 validates a new group of practitioners who are represented in the process of doing their work, their individuality exhibited along with their virtues of seriousness, industriousness, and care.

The practitioner is also a central figure in the *Feuerwerkbuch,* an anonymous treatise that was written around 1420 and widely distributed in manuscript copies before it was printed in 1529. The treatise describes itself as concerned with the guns, missiles, gunpowder, and pyrotechnics by which princes, counts, lords, knights, and cities protect themselves when besieged. These princes and cities should have "servants who are godly and steady people," who will voluntarily give their bodies, lives, and property in the fight against the enemy before surrendering. They should also be "wise people" who know how to bring up guns and missiles in an emergency, how to attack, how to build walls and bulwarks, and how to use arms effectively. Finally, they need to get along with each other, not quarrel, and settle their disputes according to the best advice.[32]

The author suggests one reason why he wrote his treatise—that technical details of gunpowder manufacture are too complex to remember without the help of writing. Yet concerns beyond the transmission of technical information are also evident. The treatise begins with twelve questions, some of which indicate an interest in causes. For example, the first question asks whether it is the fire or the force of the vapor produced by the fire that drives the stone from the cannon. The author argues that it is the vapor and proposes an experiment to prove it. Take a pound of good powder, put it in a thick wine cask, and close the cask well so that no vapor may come out. Then ignite the powder at the hole; the powder at hand is burned and the vapor destroys the cask. The second question asks whether the saltpeter or the sulfur pushes the stone. It is both. When the powder is kindled, the sulfur is hot and the saltpeter cold. Since heat won't admit cold and cold won't admit heat, they are two contradictory things that won't tolerate each other. Other questions concern technique: How much powder should be used? Should the wedges pushed against the stone ball in the barrel be made of soft wood like linden or hard wood like oak or beech? Is the gunpowder better corned or in the form of a fine powder?[33]

The *Feuerwerkbuch* also enumerates the qualities of a good gunner. "Above all he should honor and hold God before his eyes." In addition, he should exercise great care when handling guns and powder. He should "rest content with the world in which he

[32] For a reprint of the 1529 edition and a translation into High German see Hassenstein, *Feuerwerkbuch von 1420,* pp. 15–16; see pp. 41–43 for the modern German: "diener / die als from[m] unnd fest leüt seyen"; "sy durch eeren willen / ir leyb lebe[n] und güt / und was in Got ye verleyhen hat gegen iren feinden / darstrecken und wagen dürfften"; "weissleut." The author further admonishes princes and rulers that they need good smiths, bricklayers, carpenters, cobblers, and also good gunners; they must be good masters to their gunners, giving them all the supplies and equipment that they need. Kramer, *Berthold Schwarz* (cit. n. 30), pp. 52–231, includes a facsimile edition and transcription of one manuscript version of the *Feuerwerkbuch* (Freiburg, Universität Freiburg, HS. 362). See also Jähns, *Geschichte der Kriegswissenschaften,* Vol. 1 (cit. n. 1), pp. 392–408.

[33] Hassenstein, *Feuerwerkbuch von 1420,* p. 16: "And thereupon since the subjects belonging to it [gunnery] are so many, which every good gunner should know, and which a master without writing cannot remember in his mind," all the necessary details are provided in the manual. ["Und darumb wann der stuck sovil sind die darzügehöred / die ein yetlicher gütter püchsenmaister künden soll / und die ein mayster on die geschrift in seinem sinne nit gedencken kan."] The treatise provides instruction on how to treat the ingredients of gunpowder (sulfur, saltpeter, and charcoal), how to mix them together, how to load the barrel, and how to discharge the cannon in diverse situations and with various kinds of ammunition such as flaming bolts and stone balls. See also *ibid.,* pp. 16–17 and 43–45 (for modern German).

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travels" and should be a "strong, undaunted man." In war, he should "hold himself pleasantly," for such people are greatly trusted. "The master should also know how to write and read"—all the elements of the art cannot be held in the head, but they can be read in the book at hand. He should know something about purifying and distilling and about defensive operations, and he should understand weights and measures. He should teach with honesty and friendliness in both his words and his works, should be thoughtful and avoid drunkenness at all times.[34]

The author proffers advice to rulers as well. Many have found themselves besieged and haven't taken proper care for defense, nor have they brought people to them "who through their technical wisdom [*Kunst Weissheit*] advise and aid in fighting their enemies." As a result, many God-fearing princes and lords of the Holy Roman Empire have come to destruction. Thus, "the advice of a true advisor of all princes" is that they, as well as other nobles and cities, provide themselves with sober, skillful gunners to whom they give adequate provisions and supplies.[35] The *Feuerwerkbuch* addresses, then, a double readership—gunners and the princes and nobles who employ them. It provides a written text that recommends and enhances an image of learning, character, and technical competence for gunner and patron alike.

Much more study is needed of the many further manuscript treatises on weaponry and other kinds of machines and devices that proliferated in the fifteenth-century German empire. They are diverse in both their specific subject matter and their form; some are illustrated, some written in prose, others in verse. Examples include HS. 25,801, in the Germanisches Nationalmuseum in Nuremberg, which depicts various ways in which gun carriages and their aiming mechanisms (such as the elevating arch and the screw) might be constructed; the manuscript known as the "Anonymous of the Hussite Wars," which illustrates and describes machines such as mills, cranes, and aquatic devices, as well as military apparatus; and the *Mittelalterlichen Hausbuch,* from about 1480, which illustrates, among other things, cannon, weight-lifting machines, mills, a spinning wheel, a wagon train, and mining activities and machines.[36]

In the last third of the fifteenth century, several prominent gunners in the region of southwest Germany supplemented their military practice with authorship. One was Johann Formschneider, active in Nuremburg between 1460 and 1470, whose name is attached to a particular group of writings and drawings of cannon.[37] Martin Merz (d. 1501) served the

[34] *Ibid.,* pp. 18 and 47 (for modern German): "sol er got eren und vor allen dingen vor auge[n] haben"; "sich auch bescheidenlich mit der welt halten / mitt der er dann wandelt"; "ain endlicher unverzagter man sein"; "sich tröstlich halten"; "d[er] maister sol auch künden schreiben un[d] lesen."

[35] *Ibid.,* pp. 31 and 69 (for modern German): "durch der kunst weissheit / radt und hülff / sy iren feinden widerstan . . ."; "radt der getrew radtgeb allen fürsten."

[36] Nuremberg, Germanisches Nationalmuseum, HS. 25,801, "Kriegsmachinen," is an early fifteenth-century manuscript that provides explanatory text in verse. Hall, *Technological Illustrations* (cit. n. 1), includes a facsimile edition of the "Anonymous of the Hussite Wars," with a transcription and English translation of the text and a technological commentary. For the last-named text see Helmuth Th. Bossert and Willy F. Storck, eds., *Das mittelalterliche Hausbuch nach dem originale in Besitze des Fürsten von Walburg-Wolfegg-Waldsee im Auftrag des deutschen veneins für Kunstwissenschaft* (Leipzig: E. A. Seemann, 1912). For descriptive inventories of fifteenth-century German manuscripts of this kind see Hall, *Technological Illustrations,* pp. 11–25, 118–133; and Jähns, *Geschichte der Kriegswissenschaften,* Vol. 1 (cit. n. 1), pp. 243–443.

[37] Formschneider's works include Munich, Bayerische Staatsbibliothek, Cod. germ. 734, fols. 60v–71r, of which fol. 60v is signed "Johann Formschneider, büchsen meister und gutter aben teurer"; a group of similar drawings is found in fragments in Munich, Bibliothek des Deutschen Museums, MS 1949/258, and Nuremberg, Germanisches Nationalmuseum, HS. 719 (formerly Kr. 300), among others. See esp. Hall, *Technological Illustrations,* pp. 21–22, 127–129; Jähns, *Geschichte der Kriegswissenchaften,* Vol. 1, pp. 392, 411; Hassenstein, *Feuerwerkbuch von 1420* (cit. n. 31), p. 86; and Eduard A. Gessler, "Hans Formschneider von Nürnberg," in *Die deutsche Literatur des Mittelalters: Verfasserlexikon,* 4 vols., ed. Wolfgang Stammler and Karl Langosch (Berlin: Walter de Gruyter, 1933–1953), Vol. 1, col. 637.

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energetic duke Friedrich I of Siegreich (ruled 1451–1476), who reorganized his state administratively while expanding his territorial power. In 1471, apparently while in Friedrich's service, Merz wrote a treatise on the mathematics of aiming cannon. His tomb is in the parish church of Amberg. The oldest memorial of a European gunner, it is covered by a full-length relief sculpture of Merz dressed in a richly ornamented cloak. Inscriptions declare that he was "famous before others in the mathematical art of gunning" and that the dukes of Pfalz always received his heart and work and were truly served by him.[38] Another Pfalz gunner, Philipp Mönch, following Merz's example, wrote a military treatise near the end of the century and included his own portrait on the title page.[39]

A very different kind of military codex was created when the emperor Maximilian, as part of a reorganization and augmentation of the imperial army, ordered the arsenals throughout his far-flung territories to be inventoried by means of large, spectacularly illustrated books—*Zeugbücher.* For each arsenal location, cannon and other weapons were listed on one or more folios and illustrated in beautiful, hand-painted images. Other German princes created similar books by collecting images from the many technical books of the preceding century in large, hand-painted weapons books.[40] Large codices containing exquisitely rendered drawings of cannon and other machines served as new visual representations of princely power.

The many German manuscript codices within this tradition of authorship need much more study with regard to both their substantive content and their relationships to one another. The variety and richness of these books suggest the probability of diverse uses. Certainly the transmission of technical information from one practitioner to another can be included among these uses. Also evident are the value of the artillery codex in patronage exchanges and its cultural utility as a representation of the power of the prince or emperor.

[38] One of his eyes is covered with a patch, his hands hold rosary beads in prayer, and his feet rest on a cannon barrel. The sculpture displays two coats of arms, one depicting a basilisk and the other a heavy gun built into a wheeled gun carriage. The treatise is in Munich, Bayerische Staatsbibliothek, Cod. germ. 599, fols. 66r–101v. For a description see Schneider, ed., *Deutschen Handschriften,* Vol. 5, Pt. 4 (cit. n. 31), pp. 225–226. The location of the copy mentioned in earlier sources—Vienna, Bibliothek des Fürsten Liechtenstein, Cod. MS 3, 1471—has to my knowledge been unknown since the end of World War II. For Merz see Eduard A. Gessler, "Martin Merz (Mercz)," in *Deutsche Literatur des Mittelalters,* ed. Stammler and Langosch, Vol. 3, cols. 368–370; George Sarton, *Introduction to the History of Science,* Vol. 3, Pt. 2: *Science and Learning in the Fourteenth Century* (Washington, D.C.: Carnegie Institution, 1948), p. 1553; and Schmidtchen, *Bombarden, Befestigungen, Büchsenmeister* (cit. n. 30), pp. 153, 182–183. For Friedrich of Siegreich see Henny Grüneisen, "Friedrich I. der Siegreiche," in *Neue deutsche Biographie* (Berlin: Duncker & Humblot, 1961), Vol. 5, pp. 526–528.

[39] Mönch's treatise is in Heidelberg, Universitätsbibliothek, Cod. palat. germ. 126. See Eduard A. Gessler, "Philipp Mönch," in *Deutsche Literatur des Mittelalters,* ed. Stammler and Langosch, Vol. 3, col. 427; Hall, *Technological Illustrations* (cit. n. 1), pp. 23, 125; Jähns, *Geschichte der Kriegswissenschaften,* Vol. 1 (cit. n. 1), p. 271; and Hans Wegener, *Beschreibendes Verzeichnis der deutschen Bilder-Handschriften des späten Mittelalters in der Heidelberger Universitäts-Bibliothek* (Leipzig: J. J. Weber, 1927), pp. 99–101.

[40] One example is Munich, Bayerische Staatsbibliothek, Cod. icon. 222, a large vellum codex of 296 folios of hand-painted illustrations of cannon with their stone or iron balls, culverins (light field guns), and other armaments from specific sites in the empire, by Bartholomeus Freysleben, a locksmith by trade who became master gunner at Innsbruck for the emperor Maximilian I between ca. 1493 and 1509. The compilation of *Zeugbücher* was an aspect of the reorganization and strengthening of the imperial army that included the creation of a center of cannon founding and arms production and a central arsenal at Innsbruck. See especially Erich Egg, "From the Beginning to the Battle of Marignano—1515," in *Guns: An Illustrated History of Artillery,* ed. Joseph Jobé (Greenwich, Conn.: New York Graphic Society, 1971), pp. 26–31; Egg, *Tiroler Geschützguss* (cit. n. 30), pp. 49–94; Hall, *Technological Illustrations,* pp. 23, 126; and Schmidtchen, *Bombarden, Befestigungen, Büchsenmeister* (cit. n. 30), esp. pp. 172–174. Another example is the large *Kriegsbuch*—Erlangen, Universitätsbibliothek, Ms. 1390 (B26)—an illustrated encyclopedia of military drawings taken from fifteenth-century books, done by Ludwig von Eybe zum Hartenstein (the younger) (1450–1521) for Pfalzgraff Philipp bei Rhein, ca. 1500. See Hall, *Technological Illustrations,* p. 124; and Otto Pültz, *Die deutschen Handschriften der Universitätsbibliothek Erlangen,* ed. Armin Dietzel and Günther Bauer (Wiesbaden: Otto Harrassowitz, 1973), pp. 39–40.

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A careful look at particular codices suggests another point. Take, for example, a codex in the New York Public Library—Spencer MS 104. The codex retains its fifteenth-century binding, made of beautiful hand-tooled calf leather stretched over wooden boards, and its iron clasp, lock, and key. Inside are careful copies of two treatises—the *Feuerwerkbuch* and a German translation of Kyeser's *Bellifortis.* The illustrations for the latter are only partially completed, and the specific ownership history of the codex in the early modern period is unknown.[41] Yet the two texts—very different treatises, which I have treated in different sections here—were put together for readers or collectors who (the beautiful binding tells us) were far from lowborn. My point is that the tradition of authorship discussed here also had the effect of creating a readership made up of individuals who valued, collected for their libraries, and perhaps even read books on the mechanical arts.

THE HUMANIST SYNTHESIS OF *TECHNE* AND *PRAXIS:* ALBERTI, CUSANUS, VALTURIO

Unlike the books of German gunners, humanist writings on the mechanical arts have been the focus of extensive scholarship. Humanism was a broadly based cultural and educational movement that favored rhetoric and ethics over logic and philosophy. In general, humanists advocated a life of active civic participation over one of contemplation. They attempted to reinstate classical, Ciceronian Latin in the place of the "barbarisms" of medieval Latinity. Finally, they wrote histories, both ancient and contemporary, the latter often undertaken as part of their service in the courts and republics of patrons.[42] To put it another way, the humanists embraced *praxis,* that is, political and military action, and the linguistic strategies that shaped it. Some humanists joined this interest in *praxis* with a concern for *techne* or *ars,* construction in the material world, and they undertook to write about that construction.

Three key figures in the humanist synthesis of *praxis* and the constructive arts were contemporaries and probably acquaintances—Leon Battista Alberti, Nicholas Cusanus, and Roberto Valturio. Alberti (1404–1472), the illegitimate son of an exiled Florentine banker, was one of the most important humanists of the fifteenth century. Supported by ecclesiastical positions and benefices, as well as the patronage of princes, he wrote on a wide range of subjects—literary, moral, and technical, including painting, sculpture, and architecture—and was involved as well in urban reconstruction and building design.[43] The German cardinal Nicholas of Cusa (1401–1464) attended the University of Heidelberg

[41] New York Public Library, Spencer MS 104.

[42] For an introduction to humanism, including extensive further bibliography, see esp. Albert Rabil, Jr., ed., *Renaissance Humanism: Foundations, Forms, and Legacy,* 3 vols. (Philadelphia: Univ. Pennsylvania Press, 1988); Pamela O. Long, "Humanism and Science," *ibid.,* Vol. 3, pp. 486–512; and Anthony Grafton, *Defenders of the Text: The Traditions of Scholarship in an Age of Science, 1450–1800* (Cambridge, Mass.: Harvard Univ. Press, 1991).

[43] The foundational study of Alberti's life and work was Girolamo Mancini, *Vita di Leon Battista Alberti,* 2nd ed. (Florence: G. Carnesecchi, 1911). See also esp. Franco Borsi, *Leon Battista Alberti* (Milan: Electa, 1975); Cecil Grayson and Giulio C. Argan, "Leon Battista Alberti," in *Dizionario biografico degli Italiani* (Rome: Istituto della Enciclopedia Italiana, 1960), Vol. 1, pp. 702–713; and Joan [Kelly] Gadol, *Leon Battista Alberti: Universal Man of the Early Renaissance* (Chicago: Univ. Chicago Press, 1969). Recent studies include Mark Jarzombek, *On Leon Baptista Alberti: His Literary and Aesthetic Theories* (Cambridge, Mass.: MIT Press, 1989); Paolo Marolda, *Crisi e conflitto in Leon Battista Alberti* (Rome: Bonacci, 1988); and Giovanni Ponte, *Leon Battista Alberti: Umanista e scrittore* (Genoa: Tilgher-Genova, 1981). See also Susannah Foster Baxendale, "Exile in Practice: The Alberti Family In and Out of Florence, 1401–1428," *Renaissance Quarterly,* 1991, *44*:720–756; and Susannah Kerr Foster, "The Ties That Bind: Kinship Association and Marriage in the Alberti Family, 1378–1428" (Ph.D. diss., Cornell Univ., 1985). A brief, useful introduction to Alberti's life is Martin Kemp, "Introduction," in Leon Battista Alberti, *On Painting,* trans. Grayson (London: Penguin, 1991).

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before coming to Italy, where he absorbed humanist learning. He pursued an illustrious career in the papal court, often served as a diplomat between Italian and German regions, and wrote brilliantly original treatises devoted primarily to theological subjects. Roberto Valturio (1405–1475) was born in Rimini, where he received a humanist education, then worked in Rome as secretary in the papal court, and finally returned to his native city to serve the lord, Sigismund Malatesta. Unlike Alberti and Cusanus, Valturio wrote only one book, *Elenchus et index rerum militarum.* It became the most widely disseminated military treatise of the fifteenth century.[44]

Battista Alberti was a preeminent man of letters who was also deeply involved in the constructive arts. Although his complex literary production cannot be recapitulated here, two points deserve emphasis. First, in addition to a base of ecclesiastical service and benefices, Alberti's authorship was fundamentally grounded in the patronage of oligarchs and princes such as the Este of Ferrara, the Gonzaga of Mantua, the Rucellai of Florence, Sigismund Malatesta of Rimini, and the Florentine pope Tommasso Parantucello—Nicholas V. Second, Alberti himself became active in the constructive arts as a consultant and designer of building projects, such as churches and the redesign of cities.[45]

Arriving in Florence sometime after the ban against his family was lifted in 1428, Alberti was deeply influenced by Florentine developments in the visual and constructive arts. He came in time to observe the spectacular achievements of Brunelleschi: his engineering feat of raising the immense dome of the Florentine cathedral, his classically proportionate building design, and his discovery of painter's perspective. Alberti also observed the perspective paintings of Masaccio and others, the gilded bronze relief sculptures by Ghiberti on the doors of the Florentine baptistery, and Donatello's newly classical sculpture.[46] He

[44] Among his other achievements, Nicholas of Cusa contributed to the resolution of the "great schism" by arguing for the authority of councils (*De concordantia catholica,* 1433) and wrote a proposal for negotiation with the Hussites. For an introduction see J. E. Hofmann, "Nicholas Cusa," in *Dictionary of Scientific Biography,* ed. Charles C. Gillispie, 18 vols. (New York: Scribner's, 1970–1990), Vol. 3, pp. 512–516; Jasper Hopkins, *A Concise Introduction to the Philosophy of Nicholas of Cusa* (Minneapolis: Banning, 1978); and Pauline Moffitt Watts, *Nicolaus Cusanus: A Fifteenth-Century Vision of Man* (Leiden: Brill, 1982). For Valturio's life see Friedrich Klemm, "Roberto Valturio," in *Dictionary of Scientific Biography,* ed. Gillispie, Vol. 13, pp. 567–568; and Pier Luigi Bassignana, ed., *Le macchine di Valturio nei documenti dell'Archivio Storico Amma* (Turin: Umberto Allemandi, 1988), pp. 13–16.

[45] An example of Alberti's use of authorship to acquire patronage can be seen in his treatise on painting. He dedicated the Latin version of *De pictura* to Giovan Francesco Gonzaga (1407–1444), "illustrious prince of Mantua." Alberti praises the prince's kindness, glory in arms, and skill in letters. The aim of the dedication is explicit: "You could know my character and learning, and all my qualities best, if you arranged for me to join you, as I indeed desire. And I shall believe my work had not displeased you, if you decide to enrol me as a devoted member among your servants and to regard me as not one of the least." Alberti's dedication brought the desired results. Although Giovan Francesco died in 1444, his son Lodovico became an important patron of Alberti's architecture, including two major churches in Mantua—S. Sebastiano and S. Andrea. See Alberti, *On Painting,* trans. Grayson (cit. n. 43), p. 36; and Leon Battista Alberti, "De pictura," in *Opere volgari,* Vol. 3, ed. Cecil Grayson (Bari: Laterza, 1973), pp. 7–107 (for the Latin text and Alberti's Italian translation). For Alberti's architecture, in addition to works cited above, see esp. Smith, *Architecture in the Culture of Early Humanism* (cit. n. 6), pp. 19–39 (for his relationship to Brunelleschi), 98–129 (for Pienza). For Alberti's involvement in Nicholas V's Rome see Carroll William Westfall, *In This Most Perfect Paradise: Alberti, Nicholas V, and the Invention of Conscious Urban Planning in Rome, 1447–1455* (University Park: Pennsylvania State Univ. Press, 1974); and Manfredo Tafuri, " 'Cives esse non licere': The Rome of Nicholas V and Leon Battista Alberti: Elements toward a Historical Revision," *Harvard Architecture Review,* 1987, *6*:60–75.

[46] For an introduction to these developments see Frederick Hartt, *History of Italian Renaissance Art: Painting, Sculpture, Architecture,* 4th ed., revised by David G. Wilkins (New York: Abrams, 1994). On artist's perspective see Samuel Y. Edgerton, Jr., *The Renaissance Rediscovery of Linear Perspective* (New York: Basic, 1975); and John White, *The Birth and Rebirth of Pictorial Space,* 3rd ed. (Cambridge, Mass.: Harvard Univ. Press, Belknap, 1987).

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made his own significant contributions to this visual revolution, in part through technical authorship.[47]

Traditional scholarship portrayed Alberti as one who changed the status of painting and architecture from mechanical to liberal arts by making those disciplines mathematical. It emphasized those aspects of Alberti's work that concerned design and mathematical proportion and tended to disregard his interests in engineering and material construction. More recently, the traditional erasure of Alberti's grounding in material manipulation and construction has been corrected. Christine Smith has cogently argued that Alberti's admiration for Brunelleschi was based not so much on his design of proportionate buildings as on the engineering triumph represented by the dome of the Florentine cathedral and the many ingenious weight-lifting machines that he invented to build it. Similarly, John Oppel suggests that although it is usually said that Alberti "raised architecture to the status of one of the liberal arts," it is "at least as true and historically more appropriate to argue that he did just the reverse, that he brought the liberal arts down to the level of the mechanical ones." For Alberti, indeed, "the whole of architecture is based on mechanics."[48] I would say, rather, that for Alberti both painting and architecture remain firmly grounded in material construction but that they are mathematical as well and are placed emphatically within the ethical world of human society. Instead of "mere" mechanical arts, they now exist as a focal point of ethical and civic life, of material construction, and of mathematics.

Alberti produced two versions of his treatise on painting—one in Latin, dedicated to a potential patron, Giovan Francesco Gonzaga, and the other in Italian, dedicated to Brunelleschi.[49] He begins *De pictura* by insisting that he is writing not as a mathematician

[47] In addition to the treatises on painting and architecture that I discuss in greater detail, Alberti's technical writings include the following:

Elementi di pittura, probably written in Florence in the early 1430s, on geometric figures and their transposition onto foreshortened planes. See Leon Battista Alberti, "Elementi di pittura," in *Opere volgari,* ed. Grayson, Vol. 3 (cit. n. 45), pp. 109–129, 341–351; and Alessandro Gambuti, "Nuove ricerche sugli 'Elementa picturae,' " *Studi e Documenti di Architettura,* 1972, *1*:133–172.

The *De statua,* a treatise on sculpture that deals with human proportions and with a device for reproducing such dimensions in stone or clay. See Alberti, *"On Painting" and "On Sculpture": The Latin Texts of "De pictura" and "De statua,"* ed. and trans. Cecil Grayson (London: Phaidon, 1972), pp. 117–144; and Jane Andrews Aiken, "Leon Battista Alberti's System of Human Proportions," *J. Warburg Courtauld Inst.,* 1980, *43*:68–96.

The *Ludi rerum mathematicarum* (ca. 1450), on measuring distances, dimensions, and weights. See Alberti, "Ludi rerum mathematicarum," in *Opere volgari,* ed. Grayson, Vol. 3, pp. 131–173; and Luigi Vagnetti, "Considerazioni sui 'Ludi matematici,' " *Stud. Doc. Architettura,* 1972, *1*:173–259.

A treatise on the use of an instrument to survey the city of Rome—*Discriptio urbis Romae.* See Vagnetti, "La 'Discriptio urbis Romae,' uno scritto poco noto di Leon Battisti Alberti," *Quaderno,* 1968, *1*:25–78.

A now-lost work that explains how he attempted to raise a sunken Roman galley from Lake Nemi—*De navis;* and another lost book on lifting weights. See Gadol, *Leon Battista Alberti* (cit. n. 43), p. 204; Mancini, *Vita di Alberti* (cit. n. 43), pp. 281 (on *De navis*), 286–288 (on *De motibus ponderis*). On *De motibus ponderis* see esp. Alberti, *Opere volgari,* ed. Grayson, Vol. 3, p. 431.

A tract on cryptography, *De compondendis cifris,* that explains his invention of a code wheel. See Gadol, *Leon Battista Alberti,* pp. 207–211.

[48] Smith, *Architecture in the Culture of Early Humanism* (cit. n. 6), pp. 19–39; and John Oppel, "The Priority of the Architect: Alberti on Architects and Patrons," in *Patronage, Art, and Society,* ed. Kent and Simons with Eade (cit. n. 7), pp. 251–267, on pp. 251, 262. The traditional view is elaborated most importantly in the classic work: Rudolf Wittkower, *Architectural Principles in the Age of Humanism,* 4th ed. (London: Academy, 1973), esp. pp. 1–56.

[49] Alberti, *On Painting,* trans. Grayson (cit. n. 43), p. 34, where Alberti praises the great achievements in the arts carried out by Brunelleschi, as well as by "our great friend the sculptor Donatello" and by Ghiberti, Luca della Robbia, and Masaccio. Alberti describes the achievements of these men as "in no way inferior to" those of the ancients and asks that Brunelleschi correct any aspect of his treatise that seems to need amendment. For the dedication to Francesco Gonzaga see note 45, above.

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but as a painter. What he means is that "mathematicians measure the shapes and forms of things in the mind alone and divorced entirely from matter." He, on the other hand, wants to speak in terms of Minerva, about visible things. Speaking as a painter—one who does not separate mathematical and theoretical concerns from the visible world—Alberti also frequently alludes to ancient texts and the ancient art of painting. He speaks, that is, not as an ordinary but as a learned painter. *De pictura* treats the process of vision, the creation of the illusion of three-dimensional space on a flat surface by one-point perspective, and procedures of composition, light, and color. It also concerns the painter's creation of *istoria,* that is, a dramatic effect in the painting that moves the soul of the viewer. Finally, it enumerates desirable moral attributes of the painter, of which the first is that he should "be a good man, well-versed in the liberal arts."[50]

Alberti presented his architectural treatise, *De re aedificatoria,* to Pope Nicholas V around 1452, during the time when both were involved in the urban renovation of Rome.[51] In this, the most important architectural treatise of the Renaissance, Alberti separates architecture from craft practice by means of mathematics and design. Yet he also grounds it in engineering for the improvement of civic life. The architect is not a carpenter but, rather, one "who by sure and wonderful reason and method, knows both how to devise through his own mind and energy, and to realize by construction, whatever can be most beautifully fitted out for the noble needs of man, by the movement of weights and the joining and massing of bodies." Architecture provides houses, walks, swimming pools, and baths to maintain human health. It supplies vehicles, mills, timepieces, and other smaller inventions that play a vital role in everyday life. It has devised methods of drawing up vast quantities of water for many purposes, and it has produced buildings for divine worship. Further, by cutting through rock, tunneling through mountains, filling in valleys, restraining the waters of the sea and lakes, draining marshes, building ships, altering the course of rivers, dredging the mouths of rivers, and constructing harbors and bridges, the architect has not only "met the temporary needs of man, but also opened up new gateways to all the provinces of the world," allowing nations to exchange food and goods as well as experience and knowledge. Architecture moreover has provided ballistic engines and machines of war, fortresses, and other things that protect liberty.[52]

Alberti repeatedly emphasizes that architecture, civic life, and human nature are closely linked. "When you erect a wall or portico of great elegance . . . good citizens approve and express joy for their own sake, as well as for yours, because they realize that you have used your wealth to increase greatly not only your own honor and glory, but also that of your family, your descendants, and the whole city." Types of buildings should be commensurate, Alberti insists, with types of people. The patron should take account of his

[50] Kemp, "Introduction," in Alberti, *On Painting,* trans. Grayson, pp. 12, 37. The allusion to Minerva is to Cicero's *De amicitia,* where the ancient author refers to a more popular kind of knowledge; for Alberti's use of *De amicitia* 5.19 see Leon Battista Alberti, *On Painting,* rev. ed., trans. John R. Spencer (New Haven, Conn.: Yale Univ. Press, 1966), pp. 18–20. For the desirable moral attributes of the painter see Alberti, *On Painting,* trans. Grayson, p. 87. On *istoria* see esp. Jack M. Greenstein, "Alberti on Historia: A Renaissance View of the Structure of the Significance of Narrative Painting," *Viator,* 1990, *21*:273–299.

[51] Leon Battista Alberti, *L'architettura (De re aedificatoria),* ed. and trans. Giovanni Orlando, 2 vols. (Milan: Il Polifilo, 1966); and Alberti, *On the Art of Building in Ten Books,* trans. Joseph Rykwert, Neil Leach, and Robert Tavernor (Cambridge, Mass.: MIT Press, 1988).

[52] Alberti mentions several of his own inventions that have to do with rigging gangplanks of ships to prevent boarding by an enemy and other devices for sinking and burning enemy ships. He promises that he will deal with war machines at greater length elsewhere, implying, perhaps, a plan to write a treatise on the subject: Alberti, *Art of Building,* trans. Rykwert *et al.,* pp. 3–4 (quotation), 135–136 (war machines and his own military inventions); and Alberti, *L'architettura (De re aedificatoria),* ed. and trans. Orlando, Vol. 1, pp. 6–13, 386–393.

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social standing when he commissions a building because "it is a sign of a well-informed and judicious mind to plan the whole undertaking in accordance with one's position in society and the requirements of use." The variety of buildings is the result of the differences in human nature. Just as various human types, such as the wise, the skilled and practical, and the wealthy, have different functions in society, so also do different building types. Some buildings are appropriate for society as a whole, some for the foremost citizens, some for the common people. Among leading citizens, those presiding over domestic councils require different buildings than those involved in executing decisions or those occupied with accumulating wealth. More broadly, the type of government determines the architectural decisions of the ruler—tyrants must make different architectural decisions than magistrates.[53]

Architecture is an ethical discipline, and one that also involves the acquisition of knowledge. It is not a perfected set of principles but involves an ongoing process of study. It requires empirical investigation and observation, as well as the collection of advice from the past—"we shall collect, compare, and extract into our own work all the soundest and most useful advice that our learned ancestors have handed down to us in writing, and whatever principles we ourselves have noted in the very execution of their works. We shall go on to report things contrived through our own invention, by careful, painstaking investigation, things we consider to be of some future use." This reporting, Alberti repeatedly emphasizes, must be open and clear, achieved with the use of concise, understandable technical language.[54]

Although Alberti's literary and practical interests were in some ways very different from Cusanus's theological concerns, the two men shared the conviction that knowledge and practical life, including practical mathematics, were closely interrelated.[55] Cusanus's views concerning such connections are strikingly evident in his *Idiota: De sapientia, De mente, De staticis experimentis* (ca. 1450). *Idiota* contains four books, the first two of which, on wisdom, take the form of a dialogue between an unlearned man (the *idiota*) and an educated orator. The *idiota* points the way to wisdom by rejecting the learning of the orator, which is based on the authority of books. He suggests instead that wisdom can be found in the streets and the marketplace, where ordinary weighing and measuring occur. By elaborating

[53] Alberti, *Art of Building,* trans. Rykwert *et al.,* pp. 4–5, 367 n. 12 ("good citizens approve"), 37 ("judicious mind"), 92–94 (building types), 117–125 (types of government and architecture). For a discussion of Alberti's civic and moral positions, as well as his debt to Cicero's *De officiis,* see John Onians, *Bearers of Meaning: The Classical Orders in Antiquity, the Middle Ages, and the Renaissance* (Princeton, N.J.: Princeton Univ. Press, 1988), pp. 147–157.

[54] Alberti, *Art of Building,* trans. Rykwert *et al.,* p. 7; see also p. 24, where Alberti stresses inventiveness rather than slavish imitation as a desirable approach to the architectural orders. On writing about architecture, he remarks: "But since it is our desire to be as limpid, clear, and expeditious as possible in dealing with a subject otherwise knotty, awkward, and for the most part thoroughly obscure, we shall explain, as is our custom, the precise nature of our undertaking. For the very springs of our argument should be laid open, so that the discussion that follows may flow more easily." *Ibid.,* p. 7. Elsewhere (p. 155) Alberti lamented the obscurity and poor Latinity of Vitruvius's text, but, like Vitruvius, he stressed the difficulty of writing on architecture. He preferred that his own "speech seemed lucid [rather] than appeared eloquent" and believed that his writing was "in proper Latin, and in comprehensible form." Openness and clarity are values that Alberti repeatedly emphasizes. See Richard Krautheimer, "Alberti and Vitruvius," *Acts of the Twentieth International Congress of the History of Art* (Princeton, N.J.: Princeton Univ. Press, 1963), Vol. 2, pp. 42–52; and Alberti, "Ludi rerum mathematicarum" (cit. n. 47), p. 173.

[55] For the reciprocal influence of Alberti and Cusanus see Riccardo Gavagna, "Un abbinamento editoriale del '500: Vitruvio e Cusano," *Rivista Critica di Storia della Filosofia,* 1975, *30*:400–410; and Gavagna, "Cusano e Alberti a proposito del 'De architectura' di Vitruvio," *ibid.,* 1979, *34*:162–176. Cusanus's own mathematical writings include the *Transmutationes geometricae* (1445), which treats changing a circle into a square and a straight line and includes many examples from practical geometrical problems.

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upon these activities as they see them in the Roman Forum, the unlearned man leads the orator to wisdom, that is, comprehension of the divine, through mathematics. Wisdom, which all men by nature desire to know, is higher than knowledge and utterly unknowable, unspeakable by all language, unintelligible by any understanding, immeasurable by all measure. Cusanus's elaboration involves the notion that the divine involves absolute number, precision, equality, and other mathematical entities that in the human realm can only be relative.[56]

The third book, *De mente,* concerns the human mind; Cusanus argues that the word for it—*mens*—derives from the same word as *measure* (*mensur*). In this book, which is set in the Jubilee year of 1450, a philosopher seeking knowledge about the mind from ancient books in Rome is found by the orator and led to the Forum, where the *idiota* is busy making spoons in the Temple of Eternity. The unlearned man defends his "unworthy" occupation by claiming that the maker of spoons and pots is even closer to the divine than the painter and others who imitate nature. There are no spoons in nature; likewise, the divine creation was original, not imitative.[57]

Cusanus moves from this remarkable vindication of artisanal labor to the fourth book, *De staticis experimentis,* which describes a series of experimentis in practical measurement, including the determination of specific gravities and the assaying of metals. Speaking through the *idiota,* Cusanus points to many situations in which comparative measurement of the weight of various substances, including the blood and urine of sick and well men, water from different fountains, stones, and metals, could lead to a better knowledge of the nature of those substances. The *idiota* insists that the myriad results of such measurements should be written down in charts and that one result would be human benefit—for example, physicians could better diagnose and treat patients.[58] For Cusanus, divine and human mathematics are separated by a conceptual gulf that renders the first perfect and absolute, the second always imperfect and relative. Nevertheless, the two remain linked, and Cusanus demonstrates a continuing interest in both.

Around the time that Cusanus completed *De mente,* Sigismund Malatesta (1417–1468), lord of Rimini, instructed his secretary, Roberto Valturio, to write a treatise on military matters. Sigismund had acquired Rimini through his father, the famous *condottiero* Pandolfo Malatesta, who had bestowed it upon him and his brothers. At the precocious age of fourteen, Sigismund wrested it by military force from his brothers and from an uncle who was contesting the bequest. In 1433, some time after his spectacular adolescent military victory, Sigismund, lord of Rimini, was knighted by the emperor Sigismund during a stop in Rimini on his way north after the imperial coronation. Thereafter, the lord of Rimini consolidated his rule by means of an extensive building campaign that included the construction of the great fort and palace, the Rocca Malatestiana, and the radical

[56] Nicolai de Cusa, *Opera omnia,* Vol. 5: *Idiota: De sapientia, De mente, De staticis experimentis,* rev. ed., ed. Renata Steiger *et al.* from edition of Ludwig Bauer (Hamburg: Felix Meiner, 1983). There is an anonymous seventeenth-century English translation—Cusanus, *The Idiot in Four Books* (London: William Leake, 1650), from which I have paraphrased text on pp. 7–8; and more recent English translations of the third book—Nicholas de Cusa, *Idiota de Mente: The Layman: About Mind,* trans. Clyde Lee Miller (New York: Abaris, 1979)—and of the fourth book—"De staticis experimentis," in Henry Viets, "*De staticis experimentis* of Nicolaus Cusanus," *Annals of Medical History,* 1922, *4*:115–135.

[57] Cusa, *Idiota de Mente,* trans. Miller, pp. 41–45. For the importance of the dialogue for the idea of creativity see Hans Blumenberg, " 'Nachahmung der Natur': Zur Vorgeschichte der Idee des schöpferischen Menschen," *Studium Generale,* 1957, *10*:266–283. *Idiota* involves a complicated commentary on Plato's *Apology* (21a–23b), in which Socrates recognizes the wisdom of understanding his own ignorance in the face of divine wisdom, and on the *Timaeus.*

[58] Cusa, *Opera omnia,* Vol. 5: *Idiota: De staticis experimentis,* ed. Steiger *et al.* (cit. n. 56), pp. 219–242; and Viets, "*De staticis experimentis*" (cit. n. 56).

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redesign (primarily by Alberti) of the Church of San Francesco, with a facade in the form of a triumphal arch, that came to be known as the "Tempio Malatestiano" (see Figure 6).[59] The form and decorations of the Tempio Malatestiano bring together Roman antiquity, humanist learning, and Sigismund's power and rulership—affiliations that Valturio also repeatedly invokes in his military treatise.

The *Elenchus et index rerum militarium* (often cited as *De re militari*) is a humanist tour de force in which Valturio compiles a vast amount of material from ancient military literature and discusses Sigismund's own military and architectural feats as if they constituted extensions of Roman magnificence. The most widely disseminated military treatise of the fifteenth century, it exists in at least twenty-two manuscript copies and (after 1472) in numerous printed editions and translations. Yet the favor it found in its own time has not been shared by modern scholars, who have generally ignored or scorned it. Samuel Y. Edgerton, Jr., for example, suggests that the popularity of the treatise discouraged the printed publication of engineering books for fifty years. His judgment seems to be based on a model of technological progress in which the most desirable engineering books are deemed to be those that convey the latest technical information.[60] Such a judgment would not have been shared by Valturio's contemporaries, who seem to have appreciated his erudite explication of ancient writings, his ability to bring together ancient and contemporary weapons, and his brilliant success at representing the power and glory of Sigismund Malatesta, his patron and prince.

Unlike most of his multifarious ancient military sources, Valturio supplements his examination of strategy, generalship, military rituals, order of battle, recruiting, and the like with extensive treatment of weapons technology—ancient, fantastical, and modern. Manuscript copies of the treatise include numerous beautifully hand-painted illustrations of military machines, indebted in part to the illustrations of Kyeser's *Bellifortis*.[61]

[59] For Sigismund Malatesta see esp. Philip J. Jones, *The Malatesta of Rimini and the Papal State: A Political History* (Cambridge: Cambridge Univ. Press, 1974), pp. 176–239; and Mario Tabanelli, *Sigismondo Pandolfo Malatesta: Signore del medioevo e del rinascimento* (Faenza: Stabilimento Grafico Fratelli Lega, 1977). Older studies include Salvatore Sibilia, *Sigismondo Malatesta: Signore di Rimini* (Turin: G. B. Paravia, 1942); and Charles Yriarte, *Un condottiere au xv siècle Rimini: Études sur les lettres et les arts à la cour des Malatesta* (Paris: J. Rothschild, 1882). For Sigismund's building projects see especially Carla Tomasini Pietramellara and Angelo Turchini, eds., *Castel Sismondo e Sigismondo Pandolfo Malatesta* (Rimini: Ghigi, 1985); and Woods-Marsden, "Images of Castles in the Renaissance" (cit. n. 7). For the Tempio Malatestiano see Corrado Ricci, *Il Tempio Malatestiano,* with appendix by Pier Giorgio Passini (1924; rpt., Rimini: Ghigi, 1974); and Charles Hope, "The Early History of the Tempio Malatestiano," *J. Warburg Courtauld Inst.,* 1992, *55*:51–154, esp. pp. 58–59 (for the papal bulls), 91 (for Piero della Francesca), 86–87 and 118–119 (for Roberto Valturio), 95 ff. (for Alberti), and 59–62, 66–68, 94–96, and 151–154 (for Sigismund's personal involvement in the project). For Piero della Francesca's work in Rimini see Marilyn A. Lavin, *Piero della Francesca a Rimini: L'affresco nel Tempio Malatestiano* (Bologna: Nuova Alfa, 1984).

[60] Valturio began the treatise around 1450 and completed it about 1455–1460. See Roberto Valturio, *Elenchus et index rerum militarium quae singulis codicis huius voluminibus continentur ut inpromptu sint universa hoc est ne talim Rerum curiosa perleganti* (Verona: Johannes ex Verona oriundus Nicholai Cyrugie Medici Filius, 1472). For a partial facsimile reproduction of one manuscript at Turin, Archivio Storico Amma, MS. lat., cart., sec. XV, 142 cc., and useful articles concerning the illustrations see Bassignana, ed., *Macchine di Valturio* (cit. n. 44), which includes an (incomplete) list of manuscript copies and further bibliography; and Galluzzi, ed., *Prima di Leonardo* (cit. n. 1), pp. 199–201. For Edgerton's discussion see Samuel Y. Edgerton, Jr., *The Heritage of Giotto's Geometry: Art and Science on the Eve of the Scientific Revolution* (Ithaca, N.Y.: Cornell Univ. Press, 1991), pp. 145–147.

[61] The respect accorded Valturio by the Malatesta is indicated by his burial in one of the exterior niches of the Tempio Malatestiano. Most of the material concerning military technology is in book 10, which concerns military armor, weapons, machines, and other devices. The early printed edition contains eighty-two woodcuts of military machines and devices, possibly made by Matteo de' Pasti, a sculptor in the court of Sigismund; see Bassignana, ed., *Macchine di Valturio,* pp. 13–16.

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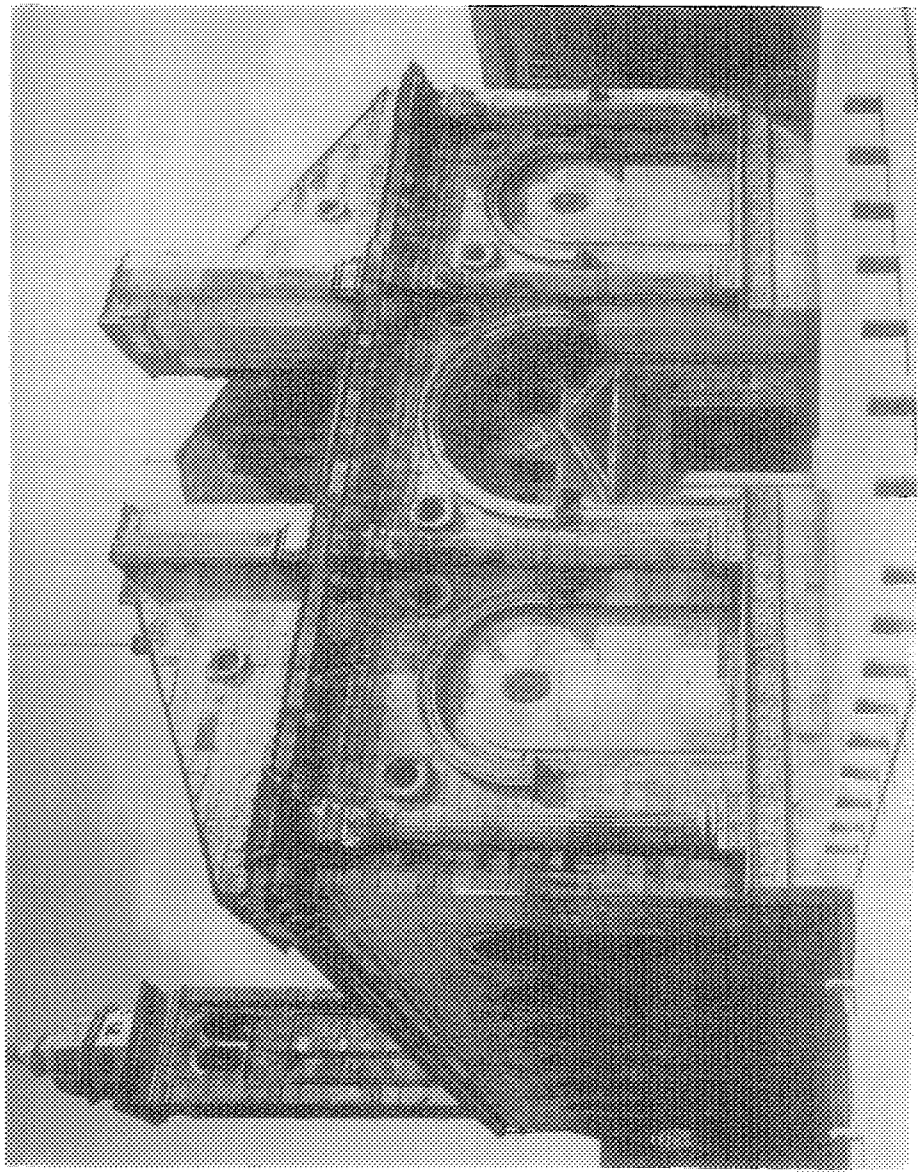


Figure 6. *Tempio Malatestiano, eastern view, Rimini, Italy. (Courtesy of Alinari/Art Resource, New York.) The lord of Rimini, Sigismund Malatesta, had the Church of San Francesco in Rimini rebuilt to his greater glory. The new facade, which resembles a Roman triumphal arch, was designed by Alberti. Some of the decorations inside were designed with the help of Roberto Valturio. Sigismund's burial niche is decorated with a fresco painted by Piero della Francesca. The fresco depicts Sigismund kneeling before a St. Sigismund who strikingly resembles the emperor of the same name. Valturio is buried in one of the outside niches.*

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Valturio suggests that by means of authorship he has placed Sigismund "in the seats of the celestial and terrestrial councils of the immortals." Just as the fame of Hector and Achilles in the Trojan War, and of many other ancients as well, has been transmitted "by the protection of letters," so his writings will serve the same function for the lord of Rimini.[62]

War itself boasts an ancient and honorable history. Emphasizing the civic nature of the military arts, Valturio suggests that military leaders develop by means of three necessary elements—nature, teaching, and practice. Also important are the instruments of war and the "many ornaments of liberal and noble arts" that are part of military science. Following the Vitruvian list of disciplines necessary to the architect, Valturio insists that the military leader must be literate, emulate the precepts of philosophy, understand history, and have good recollection. He is not ignorant of oratory and poetry; he understands music, arithmetic, geometry, the stars of the sky, the laws and rules of diverse people, and, finally, gymnastics and military practice in both peace and war.[63]

The military leader should be both "literate . . . and learned." He should be knowledgeable, not in "the vulgar, crass, and crude" things that are now used in camp, but in "noble and very precise subjects joined with knowledge of many things." Such knowledge enables the captain to read and understand the precepts and examples of those who wrote. Indeed, Valturio counsels Sigismund, when at home "you can be alone with old friends, that is, with books. Before all things, it is necessary to be soaked with ingenuity and almost seized by the work of the best teachers."[64]

Valturio provides a long list of Greek and Roman generals who were both great soldiers and learned and studious men. Primacy belongs to Alexander of Macedon, whose tutor was Aristotle. The list also includes Julius Caesar—"it is said that Julius Caesar never missed a day in the pursuit of letters"—and Augustus, an avid reader and patron of poets and scholars. Valturio moves from his last ancient example, Marcus Aurelius, directly to Sigismund himself, describing the lord of Rimini as one who reads and hears much, who has great knowledge and the seeds of all the arts, who takes pleasure in subtle questions, favors orators and poets, and collects books, which will be useful in the present and the future. Valturio praises Sigismund for his military prowess, his poetry, his buildings (he describes the palace and fort of Rimini in detail), and his growing library, which he compares to the famous libraries of antiquity. Given the vicissitudes of fortune, the impermanence of all monuments, the monument of letters lasts longest.[65]

The writings of Valturio, Cusanus, and Alberti exemplify a strain of humanism that was deeply concerned with the technical and constructive arts. Yet the three men were not artisans but university-educated humanists who stand among the most learned and influential authors of the fifteenth century. Their collective grasp of ancient learning and hu-

[62] Valturio, *Elenchus et index rerum militarium* (cit. n. 60), "Ad magnanimum . . . Sigismundu[m]," n.p.: "concilium caelestium terrestriumq[u]e insedibus i[m]mortalitatis"; "litterar[um] custodiae." Valturio adds that he hopes that he himself might also achieve immortality in future generations through his writings.

[63] *Ibid.*, bk. 1, ch. 2.: "plurima liberaliu[m] et optimarum artiu[m] orname[n]ta." (Book 1 treats the origin of war and its history in ancient times.) For the Roman architect Vitruvius's list of the disciplines that should be known by the architect see *De architectura* 1.1.3.

[64] Valturio, *Elenchus et index rerum militarium*, bk. 1, ch. 3: "litteratum . . . et eruditum"; "Non vulgarem: non crassam: et barbaram"; "sed pr[a]eclaram et accuratam illam cum rerum multarum scientia coniunctam" (what the crude, vulgar camp learning is, he does not elaborate); "cum veteribus amicis idest libris esse seorsu[m] possis: ante omnia tinctu[m] esse oportet ingeniu[m] et quasi iniciatu[m] pr[a]eceptoris optimi opera."

[65] *Ibid.*: "Nullum a studiis litterarum diem intermisisse Iul. Cae. traditur." Valturio describes the Rocca Malatestiana, including its walls, towers, and munitions, as "a true marvel of the magnificence of Italy [italic(a)e magnificenti(a)e vera admiratio]."

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manist Latinity was unparalleled in their own time. These men were not stooping to dabble in the low mechanical arts. Rather, they were participating in a reformulation of the value and place of technological culture. Valturio's studious military leader with a technological arsenal at hand; Cusanus's orator leaving his books to learn from transactions in the marketplace and from the unlearned artisan; Alberti's reformulation of the principles of painting, sculpture, and architecture in terms of rational principles, mathematics, and engineering—these were not the delusions of men whose fantasies had taken them outside of their own cultures. Neither is there the slightest reason to believe that any of the three was engaged in combating the hierarchical social structure of his time. The careers and authorship of all three were fundamentally based on patronage, whether clerical or princely. Their writings represent the highest expression of a culture in which political power and certain constructive and mechanical arts had come to form a close alliance.

PRACTITIONER AUTHORSHIP IN LATE QUATTROCENTO ITALY

Just as humanists were influenced by visual and constructive arts in the fifteenth century, artisan practitioners were inspired by humanist writings on such arts. In the second half of the century, artisan-trained practitioners such as Lorenzo Ghiberti, Antonio Averlino, Piero della Francesca, Francesco di Giorgio Martini, and Leonardo da Vinci practiced skilled crafts and also wrote treatises. By means of authorship, they aided in the transformation of certain arts from skilled practice to rational discipline and simultaneously improved their own fortunes as well. Artisans trained in workshops, such as Ghiberti, Francesco di Giorgio, and Leonardo da Vinci, struggled to learn Latin; yet they shared with the humanists a cultural arena of common interests and patronage. Practitioners and humanists together created a middle ground of discursive and constructive practice within the space that had separated learned and artisanal culture in earlier centuries.

The Florentine Lorenzo Ghiberti (1378?–1455) trained as a goldsmith in the workshop of his stepfather Bartolo di Michele, called Bartoluccio. In 1400 he established his reputation by winning a competition to make relief panels for the gilded bronze doors of the Florentine baptistery. The award, which made the Bartoluccio workshop the most important bronze foundry in the city, also provided the basis for Ghiberti's subsequent career (see Figure 7).[66] Much later, around 1447, Ghiberti wrote his tripartite treatise on sculpture known as the *Commentarii,* which includes a history of sculpture that begins in antiquity and ends with a catalogue of his own works. Like Valturio, Ghiberti excerpts extensively from ancient writings, beginning with a citation from the book on siege machines by Athenaios Mechanicos that advocates brevity and clarity over rhetorical polish. Ghiberti believed that in antiquity skilled practitioners routinely wrote books that elucidated the principles of their art but that this practice had lapsed. Only in his own age (and his own work) were ancient heights again reached in matters of art itself and in the elucidation of the principles of art in writing.[67]

[66] For Ghiberti's life and works see Richard Krautheimer with Trude Krautheimer-Hess, *Lorenzo Ghiberti,* 3rd ed. (Princeton, N.J.: Princeton Univ. Press, 1982). See also *Lorenzo Ghiberti: "Materia e rationamenti"* (Firenze, Museo dell'Accademia e Museo di San Marco, 18 Ottobre 1978–31 Gennaio 1979) (Florence: Centro Di, 1978); and *Lorenzo Ghiberti nel suo tempo,* 2 vols. (Atti del Convegno Internazionale di Studi, Firenze, 18–21 Ottobre 1978) (Florence: Leo S. Olschki, 1980).

[67] Julius von Schlosser, *Lorenzo Ghibertis Denkwürdigkeiten (I commentarii),* 2 vols. (Berlin: Julius Bard, 1912). See also Klaus Bergdolt, ed. and trans., *Der dritte Kommentar Lorenzo Ghibertis: Naturwissenschaften und Medizin in der Kunsttheorie der Frührenaissance* (Weinheim: Acta Humaniora, 1988), a commentary on and German translation of Ghiberti's third book on perspective. There is also an Italian edition in which Ghiberti's often difficult Italian has been modernized: Lorenzo Ghiberti, *I commentari,* ed. Ottovio Morisani (Naples: Riccardo Ricciardi, 1947).

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Figure 7. *Lorenzo Ghiberti's gilded bronze doors, known as the "Gates of Paradise," baptistery, Florence, Italy. (Courtesy of Alinari/Art Resource, New York.) Ghiberti's spectacular doors display ten large panels of Old Testament scenes rendered in Albertian perspective. Ghiberti included his own self-portrait within one of the medallions of the frame, placing it near his signature at about the eye level of the average viewer. The doors took more than twenty years to complete. The scenes were first modeled in wax, then cast in bronze, then gilded with gold, and finally finished. As he completed the process of construction, Ghiberti also worked on his* Commentarii.

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The difficulty and obscurity of the *Commentarii* are the evident result of Ghiberti's attempt to utilize Latin texts, a project for which his command of the language was clearly inadequate. Yet his linguistic difficulties should not obscure his shrewd choice of excerpts or his astute use of them to develop his own notion of learned practice. Ghiberti repeatedly refers to the writings (gleaned primarily from Vitruvius and Pliny) of ancient practitioners. Painting and sculpture "is a knowledge adorned with many disciplines and various teachings, which is the highest invention of all the other arts. It is made with certain meditation which is completed through material and reasoning." Things that are fabricated, Ghiberti insists, "through proportions by astuteness and by reason" can be demonstrably explained. Paraphrasing Vitruvius (*De architectura,* 1.1.2), he points to the weaknesses of both the unlettered practitioner and the unskilled learned man. Both skill (*ingegnio*) and learning (*disciplina*) are necessary to sculpture. Moreover, the sculptor must have good character traits as well. He must learn philosophy because it fills him "with [a] great soul," one that "is not arrogant" and, moreover, is "moderate and humble and faithful and without avarice." A work made without faith or chasteness cannot be perfect.[68]

The author of the first Italian vernacular architectural treatise, Antonio Averlino (ca. 1400–ca. 1462), was trained as a goldsmith in Florence, probably in the Ghiberti workshop. In 1433 Averlino, who called himself Filarete, "lover of virtue," was working in Rome, assisting in the coronation ceremonies of the emperor Sigismund. From 1433 to 1445 he was at work on the bronze relief panels for the great doors of St. Peter's. In the early 1450s, enticed by the patronage of Francesco Sforza, he moved to Milan, where he supervised the design and construction of a number of important buildings over the protests of local guildsmen. He completed his architectural treatise in the early 1460s, dedicating one copy to Francesco Sforza and another to Piero di Medici of Florence.[69]

Francesco Sforza (1401–1466) was one of the most famous *condottiere* of the fifteenth century who in 1450 became the duke of Milan through marriage, diplomatic maneuver, and military conquest. Averlino's treatise, a direct product of Sforza's patronage, is a narrative tale in which a new city, Sforzinda, is designed and constructed. The treatise provides a highly idealized account of the relationship between architect and patron, represented by Averlino and Francesco Sforza themselves. Throughout, the architect is treated as an equal, admired greatly, and paid handsomely.[70]

[68] Schlosser, *Lorenzo Ghibertis Denkwürdigkeiten,* p. 4 (1.2): "L'iscultura et pictura è scientia di più discipline et di varij amaestramenti ornati, la quale di tutte l'altre arti è somma inventione, è fabricata con certa meditatione la quale si compie per materia et ragionamenti." *Ibid.,* pp. 4–6: "per proportione d'astutia et di ragione"; "con magno animo"; "non sia arrogante"; "agevole et humile et fedele et sança avaritia."

[69] There is a modern Italian edition of the treatise and an English translation, both of which contain biographical and bibliographical information. Antonio Averlino detto il Filarete, *Trattato di architettura,* 2 vols., ed. Anna Maria Finoli and Liliana Grassi (Milan: Il Polifilo, 1972), wherein pp. lxxxviii–xci is a useful chronology of Averlino's life; for the English translation see John R. Spencer, ed. and trans., *Filarete's Treatise on Architecture,* 2 vols. (New Haven, Conn.: Yale Univ. Press, 1965). See also Angiola Maria Romanini, "Antonio Averlino (Averulino), detto Filarete," in *Dizionario biografico degli italiani* (Rome: Istituto della Enciclopedia Italiana, 1962), Vol. 4, pp. 662–667.

[70] See, e.g., Spencer, ed. and trans., *Filarete's Treatise on Architecture,* Vol. 1, p. 79, where a dinner conversation between the architect, the patron, and other nobles is described. The architect (Averlino himself) explains the many good omens that have occurred during the building of the city and reports that "my Lord and the others were so taken with love for me that they gave me enough to live honorably." For Francesco Sforza and the court of Milan under his rule see esp. Franco Catalano, *Francesco Sforza* (Milan: dall'Oglio, 1983); Giorgio Chittolini, ed., *Gli Sforza, la chiesa lombarda, la corte di Roma: Strutture e pratiche beneficiarie nel ducato di Milano (1450–1535)* (Naples: Liguori, 1989); *Gli Sforza a Milano e in Lombardia* (cit. n. 11); Gary Ianziti, *Humanistic Historiography under the Sforzas: Politics and Propaganda in Fifteenth-Century Milan* (Oxford: Clarendon, 1988); and Evelyn S. Welch, *Art and Authority in Renaissance Milan* (New Haven, Conn.: Yale Univ. Press,

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Architect and patron alike are fascinated both by design and by the technical aspects of construction. Averlino describes a dinner in which the architect, the lord, and the lord's son eat together as equals and discuss the new city. The lord's son is "so pleased by our conversation" that he falls "in love with this science of architecture" and begs his father to let the architect show him everything that had been done. The request granted, the young prince becomes the pupil of the architect, ever gaining in skill and delighted with his instruction. The prince's interests include strictly technological ones. He wants to know how metal is melted, how furnaces for melting bronze are made; he also asks for information about glass furnaces.[71]

The close association of the architect practitioner and the noble patron is paralleled by institutions in the new city in which learned disciplines and the crafts are taught side by side. Averlino proposes a school for impoverished youth that would provide instruction in letters and good habits but also in "every branch of knowledge and every skill." Instructors would include doctors of laws, of medicine, of canon law, and of rhetoric and poetry. The proposed school is unusual, however, because "some manual arts would also be taught here" by craftsmen. These include "a master of painting, a silversmith, a master of carving in marble and one for wood, a turner, an iron smith, a master of embroidery, a tailor, a pharmacist, a glassmaker and a master of clay." All the children could thereby be trained under the same roof in the discipline to which their souls and intelligences were most suited. Filarete's city also contains a "House of Virtue" in which instruction is offered in the seven liberal arts, areas are designated for military exercises, and all the crafts and trades are both practiced and taught. When the students are "judged by good masters, and have been educated in this place," they are "given the degree like the doctors." Three governors rule the temple of virtue—a doctor, a decorated soldier, and an artisan.[72]

Like Averlino, the painter Piero della Francesca (ca. 1415–1492) combined craft practice with authorship. Piero was born in Sansepolcro in central Italy and often worked there, but he also traveled to other Italian courts and cities at the behest of princely patrons. In addition to painting, Piero worked at constructive projects such as fortification and hydraulic engineering. As a painter, he was called to Perugia, Florence, Arezzo, Rome, and Rimini (where he painted a fresco, in the Tempio Malatestiano [see Figure 6], of Sigismund Malatesta kneeling before a St. Sigismund who strikingly resembles the emperor). He also worked in Urbino in the court of Federico II da Montefeltro.[73] Federico was a skilled

1996). For Francesco's father, Musio Attendolo Sforza, and his early career as a soldier see Mallett, *Mercenaries and Their Masters* (cit. n. 11), pp. 66–69 and passim; and Georges Peyronnet, "François Sforza: De condottiere à Duc de Milan," in *Gli Sforza a Milano e in Lombardia,* pp. 7–25. For the Sforza court under Francesco's son Galeazzo Maria Sforza (1444–1476) see Gregory Lubkin, *A Renaissance Court: Milan under Galeazzo Maria Sforza* (Berkeley/Los Angeles: Univ. California Press, 1994).

[71] Spencer, ed. and trans., *Filarete's Treatise on Architecture,* Vol. 1, p. 79. Filarete explicates technology as well as design. His treatise contains one of the earliest descriptions of a modern blast furnace and describes techniques such as how to make a plaster that does not stain. He mentions technical books that he is in the process of writing (none of which, if they were completed, are extant) on "agriculture," on "technical matters," and on "engines." See *ibid.,* pp. 232–233 (furnaces), 258–259 (plaster), 277 (treatise on engines), 317 (where he notes that two books of his treatise on agriculture are complete); see books 22, 23, and 24 of his treatise for his discussions of drawing, perspective, and painting. See also John R. Spencer, "Filarete's Description of a Fifteenth-Century Italian Iron Smelter at Ferriere," *Technol. Cult.,* 1963, *4*:201–206.

[72] Spencer, ed. and trans., *Filarete's Treatise on Architecture,* Vol. 1, pp. 228, 254–255.

[73] Piero was the son of a leather merchant whose family became involved in the production of the herb *guado,* which—made into dyer's indigo—was an important ingredient in the production of cloth. He owned property in Sansepolcro and seems to have enjoyed financial security based on his own and his family's holdings. For his life and works, with reference to the extant documents, see esp. Eugenio Battisti, *Piero della Francesca,* 2 vols. (Milan: Istituto Editoriale Italiano, 1971), which includes a summary of documents and sources (Vol. 2,

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military leader who reshaped his city-state by means of a massive building program and surrounded himself with courtiers distinguished in letters and the arts. He had a frieze depicting military and other machines taken from the treatises of Valturio and Francesco di Giorgio built into the facade of his palace in Urbino.[74]

In Urbino and elsewhere, Piero della Francesca created luminous, riveting paintings that reveal his fundamental grasp of mathematics and perspective.[75] Through his writings, Piero elaborated his acute understanding of mathematical principles, which was informed by his knowledge of ancient texts, especially those of Euclid. *Trattato d'abaco* (ca. 1450), concerning commercial mathematics and geometry, was written for a member of a merchant family of Sansepolcro.[76] *De prospectiva pingendi,* written for Federico II da Montefeltro in the 1470s, treats painter's perspective; and *De corporibus regularibus,* composed between 1482 and 1492 for Federico's son, Guidobaldo da Montefeltro, is devoted to the

pp. 213–246); Carlo Bertelli, *Piero della Francesca,* trans. Edward Farrelly (New Haven, Conn.: Yale Univ. Press, 1992), esp. pp. 7–50; Paolo Dal Poggetto, *Piero e Urbino, Piero e le corti rinascimentali* (Venice: Marsilio, 1992); Marilyn Aronberg Lavin, *Piero della Francesca* (New York: Abrams, 1992); and Antonio Paolucci, *Piero della Francesca: Catalogo completo dei dipinti* (Florence: Cantini, 1990). For Piero's activity as "ingegnere idraulico" see Margaret Daly Davis, *Piero della Francesca's Mathematical Treatises: The "Trattato d'abaco" and "Libellus de quinque corporibus regularibus"* (Ravenna: Longo, 1977), pp. 16–17 n. 49.

[74] Federico (1422–1482), prince of Urbino, was a brilliant soldier who led his army in the service of the pope, Venice, Florence, and the Aragonese of Naples and against his fierce rival Sigismund Malatesta of Rimini. His courtier Vespasiano da Bisticci presents him in terms of the "Valturian" soldier-scholar: "no other united as he did, in his own person the soldier and the man of letters, or knew how to make intellect augment the force of battalions." Vespasiano da Bisticci, *Renaissance Princes, Popes, and Prelates: The Vespasiano Memoirs: Lives of Illustrious Men of the Fifteenth Century,* trans. William George and Emily Waters (1926; rpt., New York: Harper & Row, Harper Torchbooks, 1963), p. 83. For Federico and his court at Urbino see esp. Giorgio Cerboni Baiardi, Giorgio Chittolini, and Piero Floriani, eds., *Federico di Montefeltro: Lo stato, Le arti, La cultura,* 3 vols. (Rome: Bulzoni, 1986); Luciano Cheles, *Lo studiolo di Urbino: Iconografia di un microcosmo principesco* (Ferrara: Franco Cosimo Panini, 1991); Walter Tommasoli, *La vita di Federico da Montefeltro (1422–1482)* (Urbino: Argalìa, 1978); and Pasquale Rotondi, *The Ducal Palace of Urbino: Its Architecture and Decoration* (London: Alec Tiranti, 1969). For the frieze on the facade of the palace see esp. Grazia Bernini Pezzini, *Il fregio dell'arte della guerra nel palazzo ducale di Urbino: Catalogo dei Rilievi* (Rome: Istituto Poligrafico e Zecca dello Stato, 1985); and Antonio Manno, "Architettura e arti meccaniche nel fregio del palazzo ducale di Urbino," in *Federico di Montefeltro,* ed. Baiardi *et al.,* Vol. 2: *Le arti,* pp. 89–104. The copy of Valturio's treatise that was in Federico's library is now Biblioteca Apostolica Vaticana, MS. urb. lat. 281.

[75] See Martin Kemp, *The Science of Art: Optical Themes in Western Art from Brunelleschi to Seurat* (New Haven, Conn.: Yale Univ. Press, 1990), pp. 27–35, for a brief, illuminating discussion of Piero's painting, in which he describes Piero as "a man deeply conversant with pure and applied mathematics, capable of writing treatises to match in quality anything produced in the Italy of his day" (p. 27). See also J. V. Field, "Mathematics and the Craft of Painting: Piero della Francesca and Perspective," in *Renaissance and Revolution,* ed. Field and James (cit. n. 5), pp. 73–95; Field, "Piero della Francesca's Treatment of Edge Distortion," *J. Warburg Courtauld Inst.,* 1986, *49*:66–90; and Marilyn Aronberg Lavin, ed., *Piero della Francesca and His Legacy* (Studies in the History of Art, 48) (Washington, D.C.: National Gallery of Art, 1995).

[76] Piero della Francesca, *Trattato d'abaco dal Codice Ashburnhamiano 280 (359*–291*) della Biblioteca Medicea Laurenziana di Firenze,* ed. Gino Arrighi (Pisa: Domus Galilaeana, 1970). This treatise was written for the Picchi family of Sansepolcro; see Battisti, *Piero della Francesca* (cit. n. 73), Vol. 1, p. 281, and Vol. 2, index, s.v. "Picchi famiglia di Sansepolcro." For the abacus tradition within which Piero worked see Richard A. Goldthwaite, "Schools and Teachers of Commercial Arithmetic in Renaissance Florence," *Journal of European Economic History,* 1972, *1*:418–433; Paul F. Grendler, *Schooling in Renaissance Italy: Literacy and Learning, 1300–1600* (Baltimore: Johns Hopkins Univ. Press, 1989), pp. 22–23, 72–73; and Warren van Egmond, *Practical Mathematics in the Italian Renaissance: A Catalogue of Italian Abbacus Manuscripts and Printed Books to 1600* (Monografia no. 4) (Florence: Istituto e Museo di Storia della Scienza, 1980). Yet Piero moves rapidly beyond the needs of commercial mathematics. He introduces innovative methods of algebraic computation, treats the measurement not of real buildings but of abstract polygons and polyhedra, handles both plane and solid geometry, and, following Euclid's *Elements,* discusses the measurements of the five regular solids and irregular solids within a sphere. See Davis, *Piero della Francesca's Mathematical Treatises* (cit. n. 73), pp. 11–20; and S. A. Jayawardene, "The 'Trattato d'Abaco' of Piero della Francesca," in *Cultural Aspects of the Italian Renaissance: Essays in Honor of Paul Oskar Kristeller,* ed. Cecil H. Clough (Manchester: Manchester Univ. Press, 1976), pp. 229–243.

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five regular solids. Piero explicated both the mathematical reasoning and the procedures required for the practice of painting, yet in range and complexity his account goes far beyond the needs of workshop practice. His prospective audience included noble patrons and learned men. Both treatises were quickly translated into Latin.[77]

Francesco di Giorgio Martini (1439–1501), from Siena, was a contemporary of Piero's who also found patronage at court in Urbino. Francesco started out as a painter and sculptor and became one of the most respected architects and military engineers of the late fifteenth century. His career was based primarily in three locations: his native city of Siena; Urbino, where he enjoyed the patronage of Federico II da Montefeltre and his successor Guidobaldo; and Naples, where he was employed by Alfonso II, duke of Calabria (1448–1495).[78]

In addition to his work as an architect and engineer, Francesco struggled to learn Latin, translated Vitruvius, and created pictorial notebooks and illustrated treatises on architecture, fortification, and military engineering. An early example of his authorship, *Opusculum de architectura,* is a book of drawings of machines and plans of fortification without descriptive text, dedicated to Federico, duke of Urbino. Reminiscent of Mariano Taccola's *De machinis,* the machine drawings—which number over two hundred—are divided into types, such as mills, pumps, siphons and waterwheels, and military devices. Francesco also authored two major treatises on architecture, engineering, and the military arts. Although the exact dates of the treatises are debated, a comparison of the earlier work to the later one reveals the striking development of his ideas and of his skills as a translator of Vitruvius.[79]

Francesco divided *Trattato I* (probably written around the late 1470s) into untitled

[77] Piero della Francesca, *De prospectiva pingendi,* ed. Giusta Nicco-Fassola, 2nd ed., ed. Eugenio Battisti *et al.* (Florence: Le Lettere, 1984); and Girolamo Mancini, "L'opera 'De corporibus regularibus' di Pietro Franceschi detto Della Francesca, usurpata da Fra Luca Pacioli," *Memorie della Rendiconti della Accademia dei Lincei: Classe di Scienze Morali, Storiche e Filologiche,* Ser. 5, 1916, *14*:441–580. The Latin translation of *De prospectiva pingendi* was made by Matteo da Borgo, who probably translated the *Libellus* on the regular solids as well; see Davis, *Piero della Francesca's Mathematical Treatises,* p. 54. Although extant copies of Piero's perspectival treatise are lacking front matter that would include a dedicatory letter, evidence that it was dedicated to Federico exists in *De corporibus regularibus,* concerning the five regular solids. The dedication letter of this later treatise, addressed to Guidobaldo, duke of Urbino, expresses Piero's hope that *De corporibus regularibus* would find a place on the shelf next to the book on perspective that he had written for Guidobaldo's father, Federico da Montefeltre. New editions of the treatises, based on additional manuscript sources, by Marisa Dalai Emiliani, Cecil Grayson, and Carlo Maccagni are forthcoming.

[78] For a comprehensive introduction to Francesco di Giorgio's career and to an extensive bibliography see Francesco Paolo Fiore and Manfredo Tafuri, eds., *Francesco di Giorgio architetto* (Milan: Electa, 1993); Gustina Scaglia, *Francesco di Giorgio: Checklist and History of Manuscripts and Drawings in Autographs and Copies from ca. 1470 to 1687 and Renewed Copies (1764–1839)* (Bethlehem, Pa.: Lehigh Univ. Press, 1992); Ralph Toledano, *Francesco di Giorgio Martini: Pittore e scultore,* trans. Massimo Parizzi (Milan: Electa, 1987); and Francesco di Giorgio Martini, *Trattati di architettura ingegneria e arte militare,* ed. Corrado Maltese, transcribed by Livia Maltese Degrassi, 2 vols. (Milan: Il Polifilo, 1967), Vol. 1, pp. xi–xxii.

[79] The dates and chronology of Francesco's notebooks and treatises are the focus of ongoing scholarly debate. See esp. Richard J. Betts, "On the Chronology of Francesco di Giorgio's Treatises: New Evidence from an Unpublished Manuscript," *Journal of the Society of Architectural Historians,* 1977, *36*:3–14; Carolyn Kolb, "The Francesco di Giorgio Material in the Zichy Codex," *J. Soc. Architect. Hist.,* 1988, *47*:132–159; and Massimo Mussini, "La trattatistica di Francesco di Giorgio: Un problema critico aperto," in *Francesco di Giorgio architetto,* ed. Fiore and Tafuri, pp. 358–359. For the *Opusculum,* some of the drawings of which derive from the books of Taccola, see Scaglia, *Francesco di Giorgio: Checklist,* pp. 43–50; for the manuscript see London, British Museum, Codex 197 B 21 [MS Harley 3281] (Scaglia, *Francesco di Giorgio: Checklist,* pp. 101–104); for the manuscript that includes the dedication to Federico see Turin, Biblioteca Reale, Codex serie militare 383 [cod. 14856; 14876; 14896 d[otazione] c[orona]; see Scaglia, *Francesco di Giorgio: Checklist,* p. 56, for Francesco's sketches on the autograph manuscript of Taccola's *De ingeneis* in Munich, Bayerische Staatsbibliothek, Codex CLM 197, part II. For Francesco's two major treatises, along with a thorough discussion of the manuscripts, see Francesco di Giorgio, *Trattati,* ed. Maltese (Vol. 1 contains *Trattato I,* Vol. 2, contains *Trattato II*).

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sections according to subject matter, such as fortresses, cities, temples, theaters, geometry, waterwheels and water mills, ways of lifting and transporting water, and military machines. *Trattato II,* although lacking a formal dedication, contains a lengthy tribute to Federico II da Montefeltre. Francesco commences this, his most mature treatise, with an introductory essay, based loosely on Pliny, that extols arithmetic and geometry. He maintains that in antiquity drawing was "reputed in the first grade of the liberal arts," rather than—as in his own time—"reputed low and inferior to many other mechanical arts." Drawing is necessary "to every human work," including invention, the explanation of concepts, and operations such as the military arts; and it is related to geometry, arithmetic, and perspective, all of which are necessary for making things "with right reason."[80]

Francesco di Giorgio's career was in many ways parallel to that of Leonardo da Vinci (1452–1519), and indeed the two men knew each other and were consulted together on at least one occasion (concerning the cathedral at Pavia). Leonardo was apprenticed in the Florentine workshop of Verrocchio in the late 1460s, at a time when the shop's numerous projects included the task of placing the heavy gilded copper sphere on top of the lantern that centered Brunelleschi's cupola. Leonardo was fascinated by this operation and by the machines employed to accomplish it, as he was also intrigued by the machines that Brunelleschi had invented to build the cupola itself. Eventually, however, he left Florence, turning to the patronage of princes, most importantly Ludovico Sforza of Milan, and others such as Cesare Borgia, *condottiero* nephew of the pope, and, finally, Francis I, king of France. These patrons valued not only Leonardo's work as a painter but also his abilities as an engineer. Leonardo's lifelong technological interests included hydraulic works, machines of all kinds, fortification, and weaponry.[81]

During most of his working life as a painter, sculptor, and engineer, especially after his arrival in Milan in the early 1480s, Leonardo created voluminous notebooks, of which approximately a third (about six thousand sheets) still exist, including the Madrid Codices, two long-lost volumes discovered in the early 1960s in the Spanish National Library. Taken as a whole, Leonardo's notebooks represent at least two kinds of authorship. Some are private notes and sketches, made in the notebooks that he habitually carried with him—Codex Madrid II is an example. Leonardo clearly intended others, such as Codex Madrid I, to be formal treatises. Leonardo planned treatises on mechanics, painting, anatomy, and hydraulics, among other things. While he was well aware of the technical writings of his contemporaries and predecessors, his own authorship was also, at least in part, motivated

[80] Francesco di Giorgio, *Trattati,* ed. Maltese, Vol. 2, pp. 293–294 (for citations), 293 n. 2 (for a discussion of the relationship of this text to Pliny's): "nel primo grado de le liberali era reputata"; "reputata vile et inferiore a molte altre arti mecaniche"; "in ogni opera umana"; "con dritta ragione." See also Vol. 2, p. 425, for Francesco's tribute to Federico. For Francesco's architectural theory, which involved anthropomorphic, modular methods of design, see esp. Henry Millon, "The Architectural Theory of Francesco di Giorgio," *Art Bull.,* 1958, *40*:257–261.

[81] The best general introduction to Leonardo's life and work is Martin Kemp, *Leonardo da Vinci: The Marvellous Works of Man and Nature* (Cambridge, Mass.: Harvard Univ. Press, 1981). For Francesco di Giorgio and Leonardo in Pavia and Milan see Pietro C. Marani, "Francesco di Giorgio a Milano e a Pavia: Consequenze e ipotesi," in *Prima di Leonardo,* ed. Galluzzi (cit. n. 1), pp. 93–104; and Gustina Scaglia, "Leonardo da Vinci e Francesco di Giorgio a Milano nel 1490," in *Leonardo e l'età della ragione,* ed. Enrico Bellone and Paolo Rossi (Milan: Scientia, 1982), pp. 225–253. For Leonardo as an engineer see esp. Paolo Galluzzi, "The Career of a Technologist," in *Leonardo da Vinci: Engineer and Architect,* ed. Galluzzi (Montreal: Montreal Museum of Fine Arts, 1987), pp. 41–109; for his interest in Brunelleschi's machines see Salvatore Di Pasquale, "Leonardo, Brunelleschi, and the Machinery of the Construction Site," *ibid.,* pp. 163–181. See also Frank D. Prager and Gustina Scaglia, *Brunelleschi: Studies of His Technology and Inventions* (Cambridge, Mass.: MIT Press, 1970), esp. pp. 65–109; and Scaglia, "Drawings of Brunelleschi's Mechanical Inventions for the Construction of the Cupola," *Marsyas,* 1960–1961, *10*:45–67.

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by patronage concerns. For example, we know from a sixteenth-century note that he began part of a book on painting (the *paragone* comparing the nobility of painting to that of sculpture) at the request of his patron Ludovico Sforza. Beyond the concerns of patronage, Leonardo was intensely involved in the world of books, as is evidenced by the list of over a hundred codices in his personal library. His own early writings were clearly inspired by Alberti's *Della pittura.*[82]

Leonardo failed to complete his treatises and presented none to patron or publisher, a circumstance that presents problems of evaluation. Unlike the writings of most other fifteenth-century technical authors, his manuscripts exist almost exclusively as autographs. Yet as Ladislao Reti pointed out, Leonardo was well aware that his unique drawings could not be reproduced by scribes, nor could woodcut print technology reproduce their fine detail. The new method of copper engraving was adequate in theory but would have been too expensive and difficult, given the large number of drawings embedded in text that he produced. Leonardo developed a whole set of illustrative techniques that allowed him to demonstrate clearly exactly how a machine, a device, or even a human body was put together. Although his inability to complete his treatises had complex causes, his awkward positioning between the cultures of scribe and print, neither of which could in his time adequately reproduce his books, clearly must have been a factor.[83]

Leonardo's Codex Madrid I constitutes a formal treatise on mechanics with two parts, one on theory and the other on practice (see Figure 8). Concerning practice, he presents often strikingly beautiful drawings of machines and mechanisms and comments on how to build them and how they work. His working methods have been elucidated by Reti, who compared several sheets of Codex Atlanticus (from one of Leonardo's private notebooks) with those of the formal treatise. Machines roughly sketched in Codex Atlanticus are redrawn with care and deliberation in Codex Madrid I. Paragraphs of text in Codex Atlanticus are crossed out and then reappear neatly inscribed (but still in mirror writing) in Codex Madrid I. Leonardo treats machines, including how to make them and how they

[82] For the complex tradition of Leonardo's notebooks see Galluzzi, "Career of a Technologist," pp. 43–47; Augusto Marinoni, "I manoscritti di Leonardo da Vinci e le loro edizioni," in *Leonardo: Saggi e ricerche,* ed. Comitato Nazionale per le Onoranze a Leonardo da Vinci nel Quinto Centenario della Nascità (Rome: Istituto Poligrafico dello Stato, 1954), pp. 231–274; Marinoni, "The Writer: Leonardo's Literary Legacy," in *The Unknown Leonardo,* ed. Ladislao Reti (New York: McGraw-Hill, 1974), pp. 56–85; and Carlo Pedretti, *Leonardo da Vinci: On Painting, a Lost Book (Libro A)* (Berkeley/Los Angeles: Univ. California Press, 1964), pp. 9 (for the note by Giovan Paolo Lomazzo), 252–259. For the Madrid Codices see Leonardo da Vinci, *The Madrid Codices,* 5 vols., ed. Reti (New York: McGraw-Hill, 1974), Vol. 2, fols. 2v–3r, Vol. 3, pp. 91–108, Vol. 5, pp. 5–8 (for Leonardo's library). Despite intense scholarship and the approximate dating of many individual sheets, Leonardo's notebooks as he wrote them cannot be fully reconstructed. Evidence that Leonardo planned formal treatises includes remarks addressed explicitly to readers and experiments with alternative tables of contents scattered throughout his notebooks; see, e.g., Jean Paul Richter, ed., *The Notebooks of Leonardo da Vinci,* 2 vols. (1883; rpt., New York: Dover, 1970), Vol. 1, pp. 13–17, 69, 190, Vol. 2, pp. 107–112, 144, 175–181, 388–389.

[83] For Leonardo's brilliantly innovative illustrative techniques see James Ackerman, "The Involvement of Artists in Renaissance Science," in *Science and the Arts in the Renaissance,* ed. John W. Shirley and F. David Hoeniger (Washington, D.C.: Folger Shakespeare Library, 1985), pp. 94–129, esp. pp. 102–111; Galluzzi, "Career of a Technologist" (cit. n. 81), pp. 96–109; and Kemp, *Leonardo da Vinci* (cit. n. 81), pp. 54–57. Francesco di Giorgio was an important predecessor in such illustration; see Martin Kemp, " 'La diminutione di ciascun piano': La rappresentazione delle forme nello spazio di Francesco di Giorgio," in *Prima di Leonardo,* ed. Galluzzi (cit. n. 1), pp. 105–112. For Leonardo's difficulty with contemporary methods of printing by woodcut and copper engraving and his invention (on paper) of relief etching see Ladislao Reti, "Leonardo da Vinci and the Graphic Arts: The Early Invention of Relief-Etching," *Burlington Magazine,* 1971, *112*:189–195. Leonardo discusses this invention in Codex Madrid II; see Leonardo, *Madrid Codices,* ed. Reti, Vol. 2, fol. 119r, Vol. 5, pp. 255–256.

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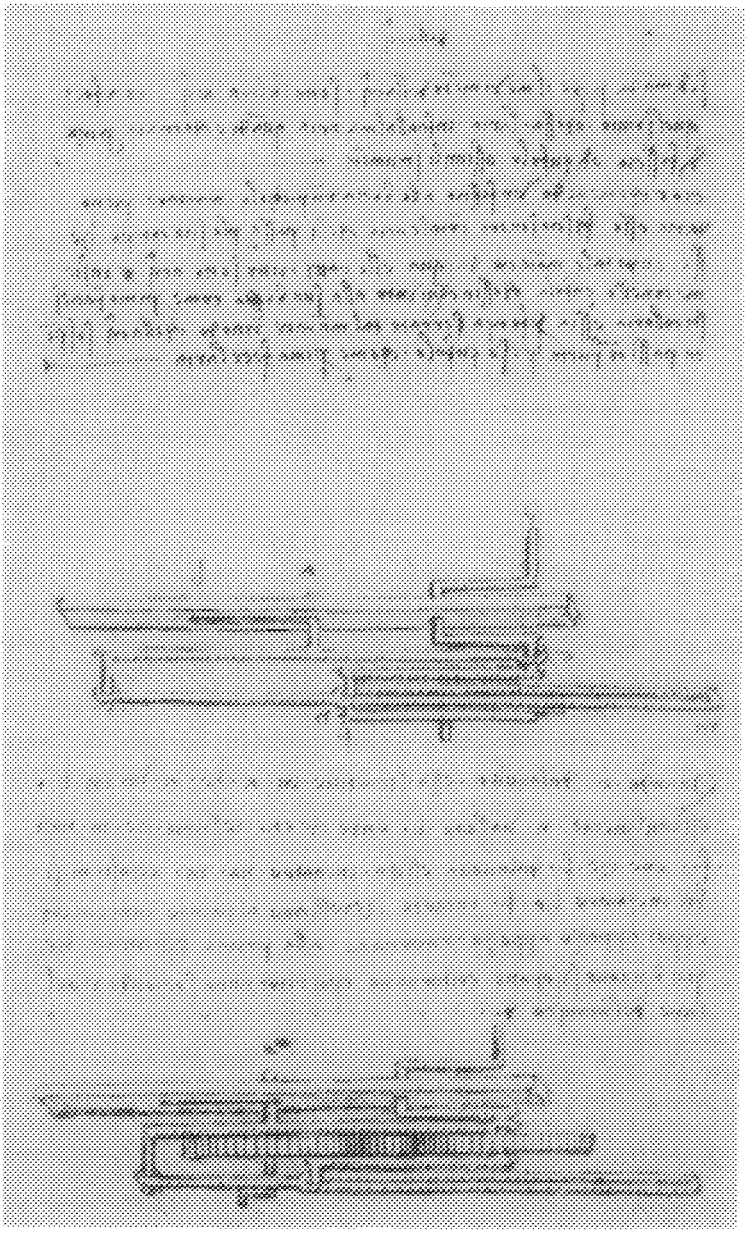


Figure 8. *Leonardo da Vinci's epicycloid gears and the notion of air and water, Madrid Codex I, fol. 111v. (Courtesy of Biblioteca Nacional, Madrid.) This sheet from Madrid Codex I is one of numerous examples in which Leonardo discusses natural and mechanical phenomena as interrelated aspects of reality that can be fruitfully examined together. Here, he asks why the motion of water forced through water (by bellows, for example) does not make a sound, whereas the motion of air forced through air (by a whistle) does make one. He then carefully describes the motion of the epicycloid gears that he has drawn. Although it is not always clear precisely what connections Leonardo is trying to make, it is abundantly evident that for him the world of nature and the world of the machine reflect one another and that analogies between the two realms have explanatory force.*

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work; he also discusses theoretical mechanics, which for him primarily concerned weight, force, impact, and motion.[84]

Paolo Galluzzi has argued that Leonardo's technical authorship was clearly different from that of predecessors such as Mariano Taccola and Francesco di Giorgio because of his profound interest in theory and his use of practical mechanics to study theoretical issues. As Galluzzi has noted, Leonardo brought together two separate traditions of the mechanical arts—one based on Aristotle's mechanics and medieval statics and kinetics, the second involving the explication of machines and the mechanical arts of the fifteenth-century engineer. My caveat is that Leonardo's predecessors included not only Mariano Taccola and Francesco di Giorgio but also Fontana, Alberti, Ghiberti, Roberto Valturio, and Piero della Francesca, among others. Leonardo's attempt to integrate theory (especially ancient and medieval theories of mechanics) with practice, as well as his use of a combination of pictorial and textual explication in his treatises, represents a development from the work of his immediate predecessors, one that arose naturally out of the rich tradition of technical authorship that preceded him.[85]

CONCLUSION

The new alliance of *praxis* and *techne* in fifteenth-century Italian and south German regions was a nexus within which authorship concerning the mechanical arts flourished. Men from diverse backgrounds—university-educated humanists and artisan-trained practitioners—took to writing about matters that had previously existed primarily in the arena of skilled practice. These authors usually wrote within the context of patronage, dedicating their books to princes, emperors, and oligarchs. Such books possessed value in relationships between patrons and clients because the arts that they treated themselves had gained cultural significance. Newly associated with learning and the liberal arts, and with the writings and heroes of antiquity, they had come to be considered intrinsic to civic, moral, and political life beyond their technical realization in the material world. The new cultural value of the constructive arts promoted the development and explication of their rational principles in writing.

The rise in status of the mechanical arts is one result of the use of building, urban redesign, and visual ornamentation to represent and promote political power in the fifteenth century. Authorship involving such arts also produced new images of practitioners—from the gunners of the *Feuerwerkbuch* to the goldsmith and architect Averlino—self-described "lover of virtue"—who embodied both skill and virtue. Nevertheless, the new cultural status of the mechanical arts, although it clearly aided the careers of certain practitioners, particularly if they wrote books, should not be taken to indicate a rise in the status of artisans in general. Whereas authorship helped to transform some arts from the arena of skilled know-how to that of discursive knowledge, it did not change artisans into learned men. It is more accurate to say that it prepared certain of the mechanical arts for appropriation by learned culture.

[84] Leonardo, *Madrid Codices,* ed. Reti, Vol. 1, fol. 18r, Vol. 4, pp. 45–48; Leonardo da Vinci, *Il Codice Atlantico di Leonardo da Vinci: Edizione in facsimile dopo il restauro dell' originale conservato nella Biblioteca Ambrosiana di Milano,* 12 vols. (Florence: Giunti Barbèra, 1973; New York: Harcourt Brace Jovanovich, Johnson Reprint, 1973), Vol. 1, fol. 77v; and Augusto Marinoni, *Il Codice Atlantico della Biblioteca Ambrosiana di Milano,* diplomatic and critical transcription, 12 vols. (Florence: Giunti Barbèra, 1975), Vol. 1, p. 57. See also Carlo Pedretti, *The Codex Atlanticus of Leonardo da Vinci: A Catalogue of Its Newly Restored Sheets,* 2 pts. (New York: Harcourt Brace Jovanovich, Johnson Reprint, 1978), Pt. 1, p. 57.

[85] Galluzzi, "Career of a Technologist" (cit. n. 81), pp. 92–95.

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The complex history of that appropriation in the sixteenth and seventeenth centuries is beyond the scope of this essay. By the phrase "from mechanical know-how to mechanical knowledge," I do not refer directly to Galilean mechanics or to the "mechanization of the world picture." Rather, I mean to suggest a change in the status of the mechanical arts accomplished in part by means of authorship—from a kind of artisanal know-how transmitted for the most part orally and utilized in the service of material construction to a group of rationalized disciplines informed by principles and explicated in books both pictorially and in writing. Without ignoring occasional prior writings on the mechanical arts, including ancient exemplars, I emphasize the significance of the great expansion of authorship concerning such arts in the fifteenth century and point to the essential context of such authorship—patronage and the use of the constructive arts for the display and legitimization of political power.

The fifteenth century is a forgotten era in the historiography of medieval and early modern science, all the more so with the burgeoning of interest in local context and meaning. Yet if the proper study of mechanics and the mechanical world view as it developed in the late sixteenth and seventeenth centuries involves the meticulous study of key texts and the careful investigation of the local cultures that inform the meanings of those texts, it should also take account of what Hans-Georg Gadamer calls the "traditionary material" that was available. Gadamer's phrase is useful because it suggests not continuous traditions, unbroken threads from the past, but cultural legacies and materials available for appropriation.[86] To put it another way, if it is methodologically sound to study "islands of discourse" in the context of local cultures (as I believe it is), it is also important to assess critically the ways in which we construct the boundaries of those discursive islands and to assess the nature of the materials from the past that are available for appropriation within those boundaries.

Fifteenth-century writings on the mechanical arts can be investigated on their own terms; they can be understood, as well, with regard to their significance for a later period. The substance of this essay concerns authorship on the mechanical arts during the fifteenth century. Yet I also suggest that such authorship had significance for the development of experimentalism and the mechanical philosophy in a later era. Scholars of the later tradition often seem to assume that seventeenth-century developments occurred against the background of a long, relatively uniform tradition of low status assigned to the mechanical arts. I hope that the present study—which, despite important differences, belongs to the tradition of scholarship represented by the late writings of Edgar Zilsel—has demonstrated the extent to which such an assumption involves an oversimplification. As a study based in the fifteenth century, it does not of course tell the complete story of transformation and appropriation that forms the background for the development of experimentalism and the mechanical philosophy. It does, however, demonstrate that the fifteenth-century proliferation of authorship in the mechanical arts forms an essential early chapter in such a story and that the history itself is far more complicated than is often assumed.

Printing greatly increased the number of technical books produced and gave far wider exposure to practitioner authors, as well as to the learned. As books became commodities, hundreds of printed technical books on artillery and fortification, mechanics, mining, metallurgy, machines, and architecture appeared, along with books of secrets and works on many other topics. Yet the positive valuation of the mechanical arts and their role in the

[86] Hans-Georg Gadamer, *Truth and Method*, 2nd rev. ed., trans. Joel Weinsheimer and Donald G. Marshall (New York: Continuum, 1993).

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legitimization of rulership were cultural phenomena that developed significantly in the century before printing. Organized textually and pictorially in manuscript books and provided with elaborated rational principles, the mechanical arts became central features of moral and political life, as they also began to take part in the construction of knowledge itself.

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